Declining Fertility and Housing Costs

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Abstract

I study how the cost of family space shapes fertility and how can housing policy affect it. Housing costs are at the center of the public debate on declining fertility: children need space, and young households, who have little savings, find that space costly to pay for. I build a life-cycle model in which households choose whether to have children, whether and how much to rent or own, and how much to save. Buying requires a down payment, and earnings are risky. I calibrate the model to the United States in 2007 and treat the fall in births since then as an exogenous decline in the taste for children. Along the transition to a new steady state with a smaller population, fewer households make housing cheaper, which feeds back into fertility. Because a child is a long commitment, earnings risk weighs on fertility, and lower risk raises births. I then use the model to assess housing policy. Easier mortgage credit could give young households access to the larger homes a child requires, but it does little for births. Policies that change long-term incentives, such as more housing supply and credit for parents, raise fertility, and their effects build up over a generation. Housing policy must therefore be judged by its long-run effects.

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1 Introduction

Fertility in the United States has fallen steadily since 2007. The period fertility rate was about 2.1 in 2007 and 1.62 in 2023. Many governments now want to raise births, and housing costs are at the center of this debate. The reason is intuitive. Children need space, and young households, who have little savings, find that space costly to pay for. Housing enters the decision to have a child in two ways: as a recurring cost, the extra rooms a family pays for every year, and, for buyers, as an upfront cost, the down payment. Since most large homes are owner-occupied, family space often requires that down payment. In the 2023 American Housing Survey, 43% of owned homes have seven or more rooms, against 7.5% of rented homes.

My goal in this paper is to ask: which housing policies can raise births, and by how much? The question looks to the future rather than the past. I take the decline in fertility as given and ask how much housing can matter for births from here on. The answer is not obvious. A child needs a discrete jump in space, and a lower down payment may be exactly what lets a young couple reach a family-size home, so easier credit could matter a great deal. Whether it does depends on how many households sit just below that threshold. Births today also become households that compete for housing a generation later, so a policy that raises births changes the price of the space the next generation needs. Weighing these forces requires a quantitative model with an equilibrium housing market. I build one and find that policy raises births when it lowers the cost of a child rather than the cost of a home, and that its effects build up over a generation.

I start with evidence from the Panel Study of Income Dynamics (Section 2). In an event study around the first birth, parents adjust both their space and their tenure. Three to four years after the birth, they live in 1.44 more rooms than two to three years before it, a 31% increase, and they are 27 percentage points more likely to own their home. Moves for more space become a third more likely after the birth, while moves for neighborhood reasons do not rise. The demand for space jumps at parenthood, and owning is the main route to a large home.

Households in my model face persistent and uninsurable earnings risk and save in a bond (Section 3). During their fertile years they decide whether to try for a child. Throughout their lives they choose whether to rent or own, how much space to occupy and how much to save. A child raises the household's needs and requires a minimum amount of space, which the household pays for in every period the child is at home. Owners can borrow against their home up to 80% of its value and renters cannot borrow, so buying requires a down payment. Rental units are capped in size, so the largest homes are only for sale. Absentee developers build housing at a cost that rises with the stock, and absentee landlords rent it out, so house prices and rents are determined in equilibrium. Children who leave home form households of their own and enter the housing market.

I calibrate the model to the United States in 2007, before the decline, to match fertility, housing and wealth moments, including the rooms parents add at the first birth (Section 4). I treat the fall in births since 2007 as an exogenous decline in the preference for children, chosen to match period

fertility in 2019–2022. The model does not explain the decline. It asks how housing responds to it and what housing policy can do from 2023 on. Every policy experiment starts from the model's 2023 economy, with the same households, wealth, homes and children, and is compared with the path the economy follows from there without the policy.

Three forces shape the results (Section 3.4). First, rent equals the user cost of owning, so a family can rent its space at about the per-period cost of owning it. Mortgage credit lowers the equity needed to buy a home, not the per-period cost of the space a child needs, which at the 2007 price is about 2.9% of mean annual earnings every year. Credit therefore raises births only where the down payment stands between a household and the home its family needs. Second, a child is a long commitment, paid for in every period, including after bad earnings draws. The utility cost of a child falls with earnings at a decreasing rate, so earnings risk raises its expected cost. In the 2007 economy, lowering the standard deviation of earnings shocks by about a quarter, with mean earnings unchanged, raises completed fertility by 12% (Section 5). Third, in a stationary equilibrium births per household are at replacement. The long-run house price is therefore the price at which households choose replacement fertility, and the number of households is the one that fills the housing stock at that price. A policy that raises fertility at a given price ends up as a higher price and more households, not as more births per household.

I then compare housing policies (Section 6). Mortgage credit for all households, a loan-to-value limit of 100% instead of 80%, raises births by 1.3% in 2023.[‡] The gain is gone within four years, and births then fall below the path without policy, by 0.6% in 2035. The completed fertility of the cohort aged 18 to 21 in 2023 falls by 1.1%. Credit moves births earlier and does not add children. The households that use it are young and buy the small homes they would otherwise rent: ownership at ages 18 to 29 rises by 21 percentage points in 2023. In the 2007 economy at a given price, credit raises the share of households aged 22 to 25 that own a home of two or four rooms from 12% to 49%, while the share owning six rooms or more, the size families choose, rises by about one percentage point. Credit lowers the cost of a home, not the cost of a child.

A housing supply expansion has a different effect. I shift the supply curve out by 4.35%, so that births in 2023 rise by about as much as under credit, 1.4%. Births then keep rising, by 3.0% by 2063 and by 4.3% in the long run, and the completed fertility of the 2023 cohort rises by 2.0%. Supply lowers the price and rents, so every cohort that enters spends its fertile years with cheaper space, and the additional children form households of their own a generation later. In the terms of demographers, credit changes the tempo of births and supply changes their quantum (Bongaarts and Feeney, 1998).

Policies that lower the cost of a child raise fertility more. Credit tied to children, a loan-to-value limit of 100% only for households with a child at home, raises births by 5.1% in 2023 and the completed fertility of the 2023 cohort by 4.9%. Insurance against low earnings raises completed fertility by 14.0%, against 7.8% for a cash transfer with the same expected value. Neither transfer is financed. The gap between them is the value of insuring the downside of earnings: a child is a

long commitment, and insurance makes it safer. A rebated property tax lowers the house price by 7.0% but raises rents by 14.5%, and births barely move. A lower price is not cheaper space.

The long run differs from the impact. Births per household return to replacement, so a policy that raises fertility shows up as more households. A 10% supply expansion leaves the long-run price unchanged and raises the number of households by 10%. Credit for parents raises fertility at any price, so fertility returns to replacement only at a price 33% higher, and at that price the housing stock holds 36% more households. Housing policy aimed at births should therefore be judged by its long-run effects, not only by its effect on impact.

Related literature. This paper sits at the intersection of several branches of the literature.

First, it relates to work on the recent fall in fertility. In the United States, the birth rate fell by about 20% between 2007 and 2020. Kearney et al. (2022) find no strong link between this decline and specific policies or economic conditions beyond the Great Recession, and point instead to shifting priorities across cohorts. Fernández-Villaverde and Norrick (2026) document that the world as a whole is likely below replacement and argue that nothing in an economy pushes fertility back to 2.1. In high-income countries, the old negative relationships between income, women's work and fertility have weakened, and the ease of combining career and family has become a central determinant of fertility (Doepke et al., 2023). I take the decline as given and ask a different question: how much housing policy can raise births from here on.

Second, the paper relates to the literature on housing costs and fertility. A large empirical literature finds that housing conditions shape childbearing. Fertility is lower where rent per room is higher (Simon and Tamura, 2009), dwelling size predicts first births (Ström, 2010), and higher house prices raise births among owners but lower them among renters (Dettling and Kearney, 2014; Lovenheim and Mumford, 2013). Access to mortgage credit also raises births (Hacamo, 2021; Cumming and Dettling, 2024; Dettling and Kearney, 2025; van Doornik et al., 2025). Two recent papers model housing and fertility jointly. Khorunzhina and Miller (2022) estimate a dynamic model of female labor supply, fertility and the first home purchase at given house prices, and Couillard (2025) estimates a model of fertility, living arrangements and location in which rents are the cost of housing. Unlike these papers, I study fertility together with saving, tenure and the choice of space under uninsurable earnings risk, in a housing market where the price is determined in equilibrium. This allows me to separate the recurring cost of family space from the equity needed to buy it, and to compare housing supply and mortgage credit within the same economy.

Third, the paper relates to work on earnings risk and fertility. Because a child is an irreversible commitment, prudent parents who face more risk want fewer children (Fraser, 2001). Quantitative life-cycle models find that uninsurable earnings risk delays and reduces births (Sommer, 2016; Ejrnæs and Jørgensen, 2020; Nakakuni, 2026), and de la Croix and Pommeret (2021) show that risk gives the postponement of a first birth an option value. Quasi-experimental studies find that job displacement and job insecurity lower fertility (Del Bono et al., 2012; Clark and Lepinteur,

2022). None of these papers includes housing. In my model, part of the commitment a child entails is the space it needs, paid for every period at a price set in the housing market. The response of fertility to earnings risk therefore depends on housing costs and on the down payment required to own family space.

Fourth, the paper builds on quantitative models of housing in macroeconomics, reviewed by Davis and Van Nieuwerburgh (2015) and Piazzesi and Schneider (2016). These models study tenure choice under earnings risk, down payments and transaction costs (Fisher and Gervais, 2011; Halket and Vasudev, 2014; Boar et al., 2022), and the joint determination of house prices, rents and ownership (Kaplan et al., 2020; Greenwald and Guren, 2025; Greaney et al., 2025).[1] In this literature, rising earnings risk accounts for much of the decline in homeownership among young households (Fisher and Gervais, 2011; Paz-Pardo, 2024). Closest to my property-tax experiment, Coven et al. (2025) show that higher property taxes lower the down payment through capitalization and raise homeownership among young households. In my model, a rebated property tax also lowers the price, but it raises the per-period cost of space, and births barely move. Household size, however, is exogenous or absent in these models. I make fertility a choice. A birth raises the household's demand for space over its life cycle and, a generation later, the number of households that compete for housing. Housing policy therefore shapes both family size and the future demand for housing.

The rest of the paper is organized as follows. Section 2 presents the evidence on housing around the first birth. Section 3 describes the model, and Section 4 its quantification and the transition from 2007 to 2023. Section 5 describes household choices in the calibrated model. Section 6 studies housing policies, and Section 7 concludes.

2 Empirical Evidence

In this Section, I show some motivating evidence on housing changes around the first birth using the Panel Study of Income Dynamics (PSID). I begin by describing changes in housing at parenthood. Three to four years after the birth, parents live in 1.4 more rooms than two to three years before it, a 31% increase, and are 27 percentage points more likely to own their home, up from 43%. Both margins, space and ownership, are central to the model of Section 3, and they crucially inform the quantification in Section 4. I then document the increased salience of space among the reasons households give for moving. One to two years after the birth, the probability of moving for more space rises by 2.1 percentage points, a 33% increase over 6.4%, while moves for neighborhood reasons do not rise. Among households that move, the share citing more space rises from 12% to 27% three to four years after the birth.

[1] In Chambers et al. (2009), mortgage innovation accounts for most of the rise in homeownership between 1994 and 2005, and in Sommer and Sullivan (2018), eliminating the mortgage interest deduction lowers house prices and raises homeownership. On the supply side, Saiz (2010) relates housing supply to land scarcity across metropolitan areas, Glaeser and Gyourko (2018) compare house prices with construction costs, and the elasticity I use is the floor-space estimate of Baum-Snow and Han (2024).

2.1 Data and Design

The PSID has followed a representative sample of U.S. families and their descendants since 1968, every year until 1997 and every two years since. I use the waves from 1968 to 2023.

In this Section, I focus on four housing outcomes. "Rooms" is the number of rooms in the home. "Owner" equals one if the household owns its home. Households that moved since the previous interview are asked why they moved, and I use the first reason they give. "Moved for space" equals one if the household moved and its first reason is expansion of housing, such as more space, and zero otherwise. "Moved for neighborhood" is defined in the same way for neighborhood reasons, such as a better neighborhood or being closer to friends or relatives; this category also includes moves to attend school. Appendix B details the variables and the sample construction.

I measure how housing changes around the first birth, relative to two to three years before it and relative to adults who never have a child. I use this reference, rather than the year before the birth, because households may move ahead of a planned birth. Formally, let Y_{it} be an outcome of adult i in survey year t and E_i the year of the first birth. I estimate

$$Y_{it} = \alpha_i + \lambda_t + x_{it}'\gamma + \sum_g \sum_{k \neq -3/-2} \delta_{g,k} \, \mathbf{1}\{E_i = g\} \, \mathbf{1}\{t - E_i \in k\} + \varepsilon_{it}, \tag{2.1}$$

where α_i and λ_t are person and survey-year fixed effects, x_{it} contains indicators for age and education, g indexes first-birth years, and k indexes two-year windows of event time $t - E_i$. The coefficient $\delta_{g,k}$ is the change in Y in window k for adults whose first birth is in year g, relative to the omitted window.

The treated group consists of adults who have a first child and who, two to three years before the birth, were already the reference person or the spouse of a household, so that their housing in the omitted window is that of their own household.[2] The control group consists of all adults who never have a child; they carry no event-time indicators. To account for effects that differ across cohorts of first births, I estimate equation (2.1) with the estimator of Sun and Abraham (2021).[3] I report

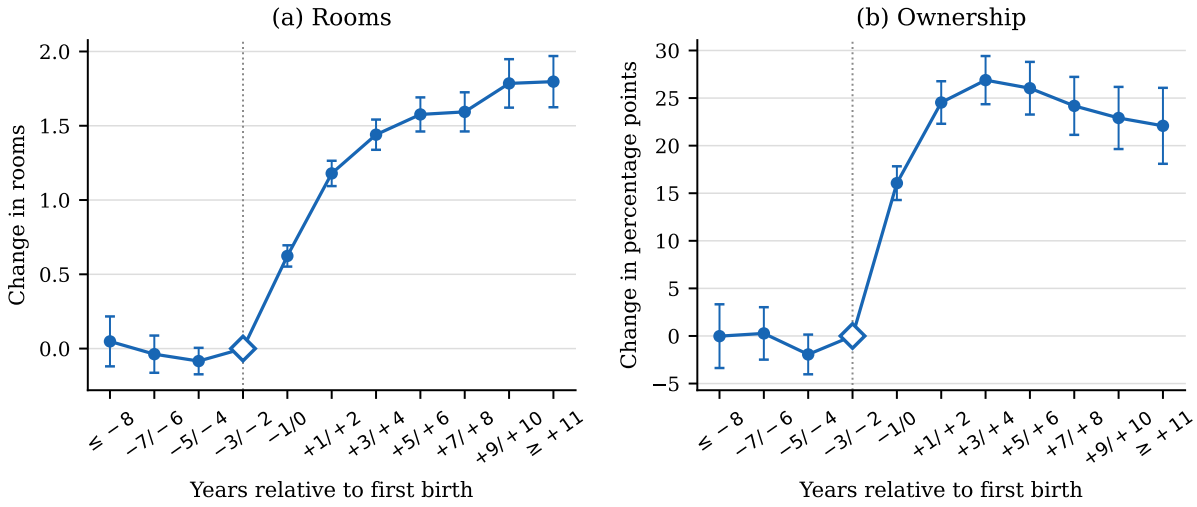
$$\hat{\beta}_k = \sum_g w_{g,k} \, \hat{\delta}_{g,k}, \tag{2.2}$$

where $w_{g,k}$ is the share of first-time parents observed in window k whose first birth is in year g. I refer to β_k as the housing response to the first birth in window k. For example, $\beta_{+3/+4}$ is the change in Y three to four years after the first birth, relative to two to three years before it and relative to the change among never-parents. Observations are weighted by the PSID longitudinal weight.

[2] The results are qualitatively similar when the sample includes, at each interview, every woman who is the reference person or the spouse of a household. For more detail, see Appendix B.4.

[3] Two-way fixed effects regressions with pooled event-time indicators are biased when the effect of the birth differs across cohorts (Goodman-Bacon, 2021). The method of Sun and Abraham (2021), followed here, estimates a separate path for each cohort of parents, compares it only with never-parents, and averages the paths with the cohort shares.

Figure 1: Rooms and Ownership Around the First Birth (PSID)



Note: This figure shows the estimates of β_k in equation (2.2), in two-year windows, with 95% confidence intervals. Rooms use the sample on which the model's event study is measured ($N = 112{,}148$); ownership uses its available observations ($N = 111{,}272$). The hollow marker is the omitted reference window.

2.2 Housing Around the First Birth

Figure 1 shows the response of housing choices around the first birth. Panel (a) shows the path of housing size. Two to three years before the birth, parents live in 4.71 rooms. Rooms rise by 0.62 in the birth window (+13%) and by 1.44 three to four years after the birth (+31%), and rise more slowly afterwards, to 1.8 more rooms eleven or more years after the birth. In the windows before the reference, the coefficients are below a tenth of a room. The adjustment is large and concentrated around the birth: two-thirds of the long-run increase occurs by one to two years after it. This indicates that, although the analysis does not rule out anticipation, children acquire particular salience around the birth itself, and the demand for space rises abruptly.

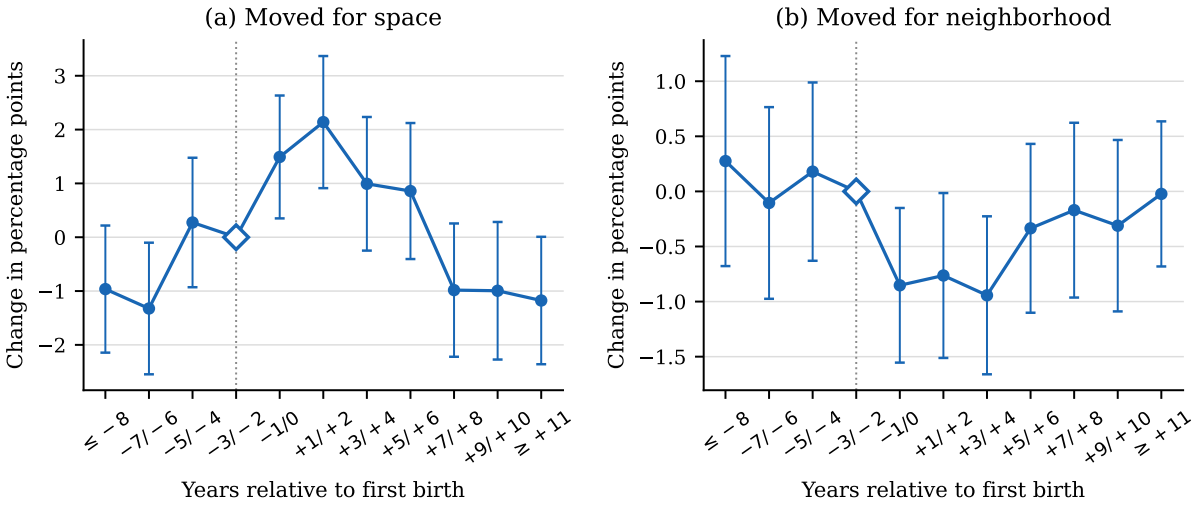
Panel (b) shows the path of homeownership. Two to three years before the birth, 43% of parents own their home. Ownership rises by 16 percentage points in the birth window (+37%) and by 27 points three to four years after the birth (+63%).[4]

Figure 2 explores the reasons households give for moving. Panel (a) shows the probability of moving for more space, and panel (b) the probability of moving for neighborhood reasons. Two to three years before the birth, 6.4% of households move for more space and 3.0% for neighborhood reasons. One to two years after the birth, the probability of moving for more space is 2.1 percentage points higher (+33%), while the probability of moving for neighborhood reasons does not rise.[5] This

[4] Appendix B.4 also shows the probability of moving since the previous interview. It is highest two to three years before the birth, at 54%, and is 17 percentage points lower one to two years after the birth and 21 points lower three to four years after it (Table 8). Couillard (2025) finds the same anticipation of births in the ACS.

[5] Among households that move, the share citing more space rises from 12% to 27% three to four years after the

Figure 2: Reasons for Moving Around the First Birth (PSID)



Note: This figure shows the estimates of β_k in equation (2.2), in percentage points, with 95% confidence intervals ($N = 116{,}697$). The outcomes are probabilities among all households, including those that did not move. The hollow marker is the omitted reference window.

highlights a strong role for concerns about size of the dwelling at childbirth.

In summary, housing is particularly relevant around childbirth, which increases the desire for size and leads people to ownership. The complex web of interactions that take place in the data motivates the writing of a model. Agents need to make a set of intertwined decisions. Having children increases desire for space, and it is easier to get this space by owning (see Appendix A). Owning requires saving to be able to afford a down payment, and children also increase general expenses. The model developed in Section 3 tries to model the interaction between these forces, in order to discipline their relative strength and potential for policy. The evidence detailed above not only motivates, but also informs my modeling choices. Because the demand for space rises at the first birth, fertility and housing are chosen jointly: a family needs a minimum amount of space before additional rooms raise its utility. In the U.S. housing stock, large homes are mostly owned rather than rented (Appendix A). The model captures this by capping the size of rental units, so that a family that wants a large home must buy it and make a down payment; the wealth of young households then matters for when they have children, and childbirth affects the desire to own, as in the data.

birth (Table 8). A larger home could also come from a move to a cheaper location rather than from a larger outlay on housing. Among households that already owned two to three years before the birth, house value rises by 30% three to four years after the birth and value per room by 11%, and moves across state lines become less frequent (Appendix B.5).

3 Model

In this Section, I develop a model of fertility and housing decisions. Households live a finite number of periods in an overlapping-generations economy. They face persistent idiosyncratic earnings risk and transitory taste shocks over fertility and housing. They derive utility from consumption, housing services and children at home, and they save in a one-period bond with an exogenous interest rate. They either rent or own their home. Owning gives more services than renting the same space, and the largest homes are only for sale. Owners can borrow against their home, up to a fraction of its value, while renters cannot borrow. A buyer must therefore make a down payment, and a seller pays a transaction cost. During a fertile period of their lives, households choose whether to try for a child. Children raise the household's needs and the value of space, so the decision to have a child is also a decision about housing. Households work until retirement and then receive a pension, financed by a payroll tax on workers, who supply labor inelastically. House prices and rents are determined in equilibrium.

3.1 The economy

I begin by describing households and their preferences, the income process, housing, the assets available for trade and the fertility process. I then describe the housing market and firms. Time is discrete and indexed by t.

Households and Preferences. The economy is populated by overlapping generations of households, with adult age $a = 0, \ldots, J$. Households can have children until age a_F and work until age a_R, after which they retire. A household of age a survives to age $a+1$ with probability s_a. Figure 3 illustrates the different periods of the life cycle. Households differ in wealth, housing, labor efficiency, age and the number of children. During their fertile years, households can choose whether to try for a child. Throughout their life, they choose whether to rent or own their dwelling, how much housing to demand and how much to save.

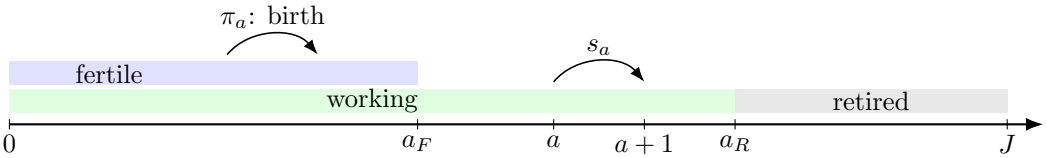


Figure 3: Life cycle of a household. Until age a_F, a try for a child succeeds with probability π_a, which declines with age. A household of age a survives to age $a+1$ with probability s_a.

Households value consumption c_t, housing services $\hat{h}_t$ and the number of children at home n_t, and

discount the future by β. A household either owns h_t^O or rents h_t^R, and its housing services are

$$\hat{h}_t = \begin{cases} \chi\, h_t^O, & \text{owner}, \\ h_t^R, & \text{renter}, \end{cases} \qquad \chi > 1.$$

The parameter χ determines the additional services a household obtains from owning a home of a given size. Flow utility is

$$u(c_t, \hat{h}_t, n_t) = \frac{1}{1-\sigma}\left[\frac{c_t^{\alpha_0}\big(\hat{h}_t - \chi^{\mathbf{1}\{\text{own}\}}\,\bar{h}(n_t)\big)^{1-\alpha_0}}{e(n_t)}\right]^{1-\sigma} + \psi_t\, v(n_t), \tag{3.1}$$

with

$$\bar{h}(n_t) = h_P\,\mathbf{1}\{n_t > 0\}, \qquad v(n_t) = n_t^{1-\gamma}.$$

The first term is utility from a Cobb–Douglas composite of consumption and housing services above a requirement $\bar{h}(n_t)$, per adult equivalent, where α_0 is the weight on consumption and σ the curvature. Children raise the household's needs in two ways. First, a household with children at home must occupy h_P units of housing before additional housing yields services; an owner obtains $\chi(h_t^O - h_P)$ and a renter $h_t^R - h_P$. The requirement does not depend on the number of children, so the first child activates it, and the household pays for it every period its children are at home, whether it rents or owns. Second, the equivalence scale $e(n_t)$, with $e(0) = 1$, measures adult equivalents relative to a childless couple.[6] The second term, $\psi_t v(n_t)$, is the benefit of children at home: ψ_t is the preference for children and γ governs how fast the benefit of an additional child declines. The first child also carries a one-time utility cost ξ at birth.

Bequests. A household that dies at the end of period t leaves the estate $x_{t+1} = b_{t+1} + (1-\mu)P_t h_t^O$, its bond position and the value of its home net of the selling cost. As in De Nardi (2004), it values the estate according to

$$B(x_{t+1}) = \theta_0\,\frac{(\theta_1 + x_{t+1})^{1-\sigma}}{1-\sigma},$$

where θ_0 scales the strength of the bequest motive and θ_1 determines the extent to which bequests are a luxury good. Estates are collected by the government.

Income. Before retirement, a household supplies one unit of labor inelastically, with efficiency $e_a z_t$. The term e_a is the age profile of earnings, and z_t is idiosyncratic labor efficiency, whose logarithm follows a first-order autoregressive process,

$$\log z_{t+1} = \rho_z \log z_t + \epsilon_{t+1}, \qquad \epsilon_{t+1} \sim N(0, \sigma_z^2),$$

[6] I use the scale of Scholz et al. (2006), $e(n_t) = \big((2+\omega n_t)/2\big)^{\nu}$, where ω is the weight of a child relative to an adult and ν governs economies of scale.

where ρ_z is the persistence and σ_z^2 the variance of the innovations. Earnings are taxed at rate τ^w, and retirees receive a pension m_t:

$$y_t(a, z_t) = \begin{cases} (1 - \tau^w)\, w\, e_a z_t, & a < a_R, \\ m_t, & a \geq a_R. \end{cases}$$

Every household also receives a transfer T_t, the rebate of property-tax revenue.

Housing. A household enters period t with the home it owned in the previous period, h_{t-1}^O, which is zero for a renter. Owner-occupied homes come in a finite set of sizes, $h_t^O \in \mathcal{H}$, and trade at price P_t per unit of space. Owners pay the property tax τ^p on the value of the home they choose, $P_t h_t^O$, and maintenance δ_H on the value of the home they enter the period with, $P_t h_{t-1}^O$. Selling a home costs a fraction μ of its value. Renters choose any size up to a cap, $h_t^R \leq \bar{h}^R$, at rent r_t per unit. Since $\bar{h}^R < \max \mathcal{H}$, the largest homes are only for sale.

Borrowing. Households save or borrow in a one-period bond, traded with the rest of the world at the exogenous gross return R_b. Let b_{t+1} denote the bond position a household chooses in period t. Borrowing must be secured by a home, so a renter cannot borrow, $b_{t+1} \geq 0$. An owner can borrow up to a fraction ϕ of the current value of its home:

$$b_{t+1} \geq -\phi\, P_t h_t^O. \tag{3.2}$$

The limit applies both to a household that buys a home and to one that stays in the home it owns. A buyer must therefore pay at least $(1 - \phi) P_t h_t^O$ itself, out of its bond position, the income of the period and the proceeds from selling any home it owned. This down payment ties ownership, and with it access to the largest homes, to the household's wealth. Debt is rolled over each period at the return R_b: mortgages have no amortization schedule and no limit on payments relative to income.

Fertility. Until age a_F, a household decides each period whether to try to have a child. A try succeeds with probability π_a, which declines with age, and a success adds one child at home in the same period. Children do not work and make no decisions; they affect the household through its needs and through the benefit of children in preferences. Each child at home leaves with probability λ per period, independently of its siblings, so a child stays at home for $1/\lambda$ periods on average. Children who leave form new households and enter the economy at age 0, with an initial wealth position drawn from an exogenous distribution.

Housing supply. Housing is built by absentee developers, who sell it to households and to landlords at the price P_t per unit. Developers turn the consumption good into housing, and the cost of a unit rises with the aggregate stock, $c_t = (H_t / H_0)^{1/\eta}$, which each developer takes as given. The rising cost stands in for land and regulation. Construction is irreversible, $I_t \geq 0$, and the stock

evolves as $H_t = (1 - \delta_H)H_{t-1} + I_t$. Developers enter freely, so the price equals the cost when they build and is below it when they do not. Together,

$$H_t = \max\{(1 - \delta_H)H_{t-1},\ H_0\, P_t^{\eta}\}, \tag{3.3}$$

where $H_0 > 0$ sets the scale of the stock and $\eta \geq 0$ is the price elasticity of housing supply. When demand rises, developers build until the price equals the cost, so the stock is on the curve $H_0 P_t^{\eta}$. When demand falls faster than the stock wears out, nothing is built, the stock shrinks at the rate δ_H, and the price falls below the cost. The elasticity governs how a change in the demand for housing is split between the price and the quantity. When $\eta = 0$ the stock is fixed and the price absorbs the whole change; the larger is η, the more the stock adjusts and the less the price moves. In a stationary equilibrium construction replaces depreciation and the stock is on the curve $H = H_0 P^{\eta}$. Developers earn no profit. The stock is divided between owner-occupied and rental housing by households' tenure choices.

Rental market. A household that does not own a home rents from landlords. Landlords are competitive and absentee: they are owned by agents outside the economy and do not build. A landlord buys housing at the price P_t, rents it to households at the rent r_t per unit, pays the property tax on its value, $\tau^p P_t$, and in the next period sells what survives, $1 - \delta_H$ per unit, at the price P_{t+1}. Landlords finance their purchases at the gross return R_b and, unlike households, pay no transaction cost when they sell. Competition drives their profit to zero, so

$$r_t = (1 + \tau^p)\, P_t - \frac{(1 - \delta_H)\, P_{t+1}}{R_b}. \tag{3.4}$$

Landlords supply any quantity of rental housing at this rent. The rent equals the user cost of housing: the price and the property tax, net of the discounted value of what survives to the next period. At a constant price, $r = \big(1 + \tau^p - (1 - \delta_H)/R_b\big)P$. The rent therefore depends on the expected path of the price, not only on its current level. When the price is expected to fall, the rent rises relative to the price, because landlords must be compensated for the expected capital loss.

Firms. A representative firm produces the consumption good, which is the numeraire, with a linear technology in effective labor,

$$Y_t = A\, L_t, \qquad L_t = \int_{a<a_R} e_a z_t \,\mathrm{d}G_t,$$

where A is productivity, G_t is the measure of households over states and L_t is the total efficiency units of labor supplied by households of working age. The firm is competitive and pays each efficiency unit its marginal product, $w = A$. The wage is therefore constant over time. Developers use the good, not labor, so construction does not change earnings; the goods market clears through net exports.

Government. The government runs a pay-as-you-go pension system, taxes housing and collects estates. It taxes earnings at the constant rate τ^w and pays the same pension m_t to every retired household. The pension system balances its budget every period, so the pension equals payroll-tax revenue divided by the number of retired households. When the number of retirees rises relative to effective labor, the pension falls. The government also levies the property tax τ^p on the value of all housing, owned and rented, and returns the revenue to households as a lump-sum transfer T_t, equal across households. Finally, the government collects the estates of the households that die. It uses them to pay new households their initial wealth and spends the remainder on government consumption, which does not affect households.

3.2 The household's problem

I next characterize the household's problem recursively. A period has two stages, shown in Figure 4. At the beginning of the period, the household observes its fertility taste shocks and decides whether to try for a child. After the birth outcome, it observes its housing taste shocks and chooses its home and saving. A birth therefore changes the demand for housing in the same period. The state of a household is its bond position b_t, the home it owns h^O_{t-1}, its efficiency z_t, its age a_t and the number of children at home n_t.

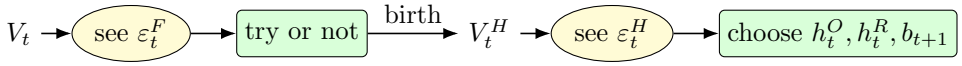


Figure 4: Timing within period t.

Fertility. Let $V_t(b_t, h^O_{t-1}, z_t, a_t, n_t)$ denote the value of a household at the beginning of period t, before it observes its fertility shocks. A household of fertile age chooses between not trying, $j = 0$, and trying, $j = A$. Each option receives a taste shock $\varepsilon^F_{t,j} \overset{iid}{\sim} \text{Gumbel}(0, \kappa)$, with $\kappa = \kappa_1$ before the first child and $\kappa = \kappa_C$ after. The separate scales allow the decision to become a parent to respond differently from the decision to have another child. The value of the household and the probability of trying are

$$\begin{aligned} V_t &= \mathbb{E}_{\varepsilon^F_{t,j}} \max_{j \in \{0,A\}} \left\{ V^F_{t,j} + \varepsilon^F_{t,j} \right\} = \kappa \log\left[\exp\left(V^F_{t,0}/\kappa\right) + \exp\left(V^F_{t,A}/\kappa\right) \right], \\ \pi^F_{t,a} &= \frac{\exp\left(V^F_{t,A}/\kappa\right)}{\exp\left(V^F_{t,0}/\kappa\right) + \exp\left(V^F_{t,A}/\kappa\right)}, \end{aligned} \tag{3.5}$$

where $V^F_{t,0}$ and $V^F_{t,A}$ are the values of not trying and trying before the shocks. Both depend on V^H_t, the value of the household after the birth outcome, which I define below:

$$\begin{aligned} V^F_{t,0} &= V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t), \\ V^F_{t,A} &= \pi_a \left[V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t + 1) - \xi \, \mathbf{1}\{\text{first child}\} \right] + (1 - \pi_a) \, V^F_{t,0}. \end{aligned}$$

Not trying leaves the number of children at home unchanged. Trying succeeds with probability π_a, adds a child at home and, for the first child, carries the utility cost ξ; if it fails, the household is in the same position as if it had not tried. Given its state, a household has a child with probability $\pi_a \pi^F_{t,a}$. Outside the fertile ages, there is no fertility choice and $V_t = V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_t)$.

Housing and savings. After the birth outcome, the household observes its housing taste shocks and chooses its home, its tenure and its bond position. Let n_{t+1} denote the number of children at home after this period's birth outcome. Before the housing shocks, the value of the household is

$$V^H_t(b_t, h^O_{t-1}, z_t, a_t, n_{t+1}) = \mathbb{E}_{\varepsilon^H_{t,j}} \max_{h^O_t, h^R_t, b_{t+1}} \Big\{ u(c_t, \hat{h}_t, n_{t+1}) + \varepsilon^H_{t,j} + \beta\, \mathbb{E}_t \Big[s_a V_{t+1}(b_{t+1}, h^O_t, z_{t+1}, a_t + 1, n_{t+1}) + (1 - s_a) B(x_{t+1}) \Big] \Big\}, \quad (3.6)$$

subject to the budget constraint of the chosen option,

$$\begin{aligned} &\text{Buy:} && c_t + b_{t+1} + \tau^p P_t h^O_t = R_b b_t + y_t + T_t - P_t \big[h^O_t - (1 - \delta_H - \mu) h^O_{t-1} \big], \\ &\text{Stay:} && c_t + b_{t+1} + (\delta_H + \tau^p) P_t h^O_{t-1} = R_b b_t + y_t + T_t, \\ &\text{Rent:} && c_t + b_{t+1} + r_t h^R_t = R_b b_t + y_t + T_t + (1 - \delta_H - \mu) P_t h^O_{t-1}, \end{aligned} \quad (3.7)$$

and the borrowing limit (3.2) for buyers and stayers, or $b_{t+1} \geq 0$ for renters. A buyer sells the home it owns, if any, and buys a new one; a stayer keeps its home, $h^O_t = h^O_{t-1}$; a renter sells the home it owns, if any, and rents. The home a household enters the period with has depreciated, so a seller receives $(1 - \delta_H - \mu) P_t$ per unit and a stayer pays maintenance $\delta_H P_t$ per unit to restore it. The expectation $\mathbb{E}_t$ is over next period's efficiency and the departure of children. The shocks $\varepsilon^H_{t,j} \overset{iid}{\sim} \text{Gumbel}(0, \kappa_T)$ are one per option j: renting, or owning each $h^O \in \mathcal{H}$. Let V^j_t denote the value of option j before the shock. Then

$$V^H_t = \kappa_T \log \sum_j \exp\big(V^j_t / \kappa_T\big), \qquad \pi^H_t(j) = \frac{\exp(V^j_t / \kappa_T)}{\sum_{j'} \exp(V^{j'}_t / \kappa_T)}, \quad (3.8)$$

where $\pi^H_t(j)$ is the probability of choosing option j. An option that leaves no saving choice within the budget and borrowing constraints cannot be chosen. Housing costs, credit and earnings risk shape V^H_t, and through it the gap $V^F_{t,A} - V^F_{t,0}$ that governs fertility.

3.3 Equilibrium

Definition 1. Given an initial measure of households over states G_0 and an initial housing stock H_{-1}, an equilibrium is a path of prices $\{P_t, r_t\}$, housing stocks $\{H_t\}$, pensions $\{m_t\}$, rebates $\{T_t\}$, value and policy functions, and measures $\{G_t\}$ consistent with the policy functions, the shocks, survival and entry, such that every period:

1. *Households* solve their problem, given prices, transfers and shocks, and rent satisfies the landlords' zero-profit condition (3.4).
2. *Housing market* clears: owned and rented housing equal the stock (3.3),

$$\int \left(h_t^O + h_t^R\right) \mathrm{d}G_t = H_t.$$

3. *Pensions*: payroll taxes pay the pensions of the N_t^R retired households, $\tau^w w L_t = m_t N_t^R$.
4. *Rebate*: property-tax revenue is paid to all N_t households, $T_t N_t = \tau^p P_t \int \left(h_t^O + h_t^R\right) \mathrm{d}G_t$.
5. *Population*: births are $B_t = \int \pi_a \pi_{t,a}^F \, \mathrm{d}G_t$, and new households are $E_t = \lambda K_t$, where K_t is the number of children at home.

Definition 2. A stationary equilibrium is a price P, a rent r, a pension m, a rebate T, value and policy functions, and a measure of households over states G, such that:

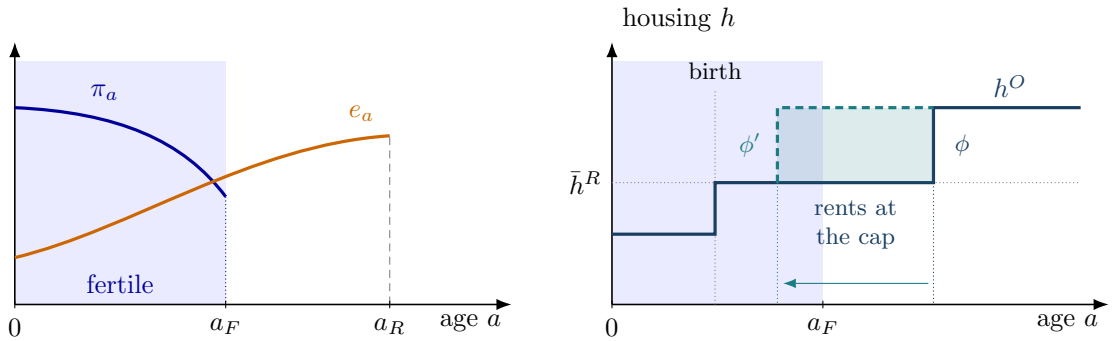
1. *Households* solve their problem, given P, r, m and T.
2. *Rent* satisfies the landlords' zero-profit condition at a constant price, $r = \left(1+\tau^p-(1-\delta_H)/R_b\right)P$.
3. *Housing market* clears, $\int \left(h^O+h^R\right) \mathrm{d}G = H_0 P^\eta$, and construction $\delta_H H_0 P^\eta$ replaces depreciation.
4. *Pensions and rebate* balance, $\tau^w w L = m N^R$ and $TN = \tau^p P \int \left(h^O + h^R\right) \mathrm{d}G$.
5. *The measure G is invariant*: the policy functions, the shocks, survival and the entry of $E = \lambda K$ new households map G into itself.

3.4 Model analysis

In this Section, I discuss the main tensions that shape fertility in the model. The first is timing: children come early in life, while earnings and wealth come later, and owning the space a child requires takes a down payment. The second is the interaction of the cost of a child's space with earnings risk. The third is the response of housing costs to fertility itself. A household that wants a child must therefore pay for its space, often before its earnings rise, and commit to paying for it while those earnings are uncertain. These tensions are then measured and assessed in Sections 4 and 5.

Borrowing limits and the timing of children. The borrowing limit ties the timing of a child to the household's wealth. A try for a child succeeds with probability π_a, which declines with age, while earnings $e_a z_t$ rise with age through e_a (Figure 5a). A child raises by $r_t h_P$ the spending needed for the same housing services, today and, with probability $1-\lambda$, next period, while earnings are still low. The home a family wants also requires a down payment, but a renter cannot borrow and an owner can borrow only against its home. Let $\zeta_t \geq 0$ denote the multiplier on the borrowing limit, $b_{t+1} \geq -\phi P_t h_t^O$ with $h_t^O = 0$ for a renter: the value, in utility, of borrowing one more unit in

period t.[7] A birth raises the multiplier when it raises marginal utility more today than next period, as for a household whose earnings rise. Such a household can delay the child, at the cost of a lower π_a. It can have the child in rented space and forgo χ and the homes larger than $\bar{h}^R$. Or it can have the child and buy once its savings cover the down payment.



(a) Fertility declines with age, earnings rise

(b) A higher $\phi' > \phi$ brings family space forward

Figure 5: Timing of fertility, earnings and the purchase of a family home

Notes: Schematic. The shaded area marks the fertile ages. In panel (b), the solid line is the housing of a household that rents until its savings cover the down payment on a family home h^O at the loan-to-value ratio ϕ. The dashed line is the same household at $\phi' > \phi$, and the shaded area is the family space it gains.

A relaxation of the borrowing limit shows how much the down payment matters for fertility. A higher loan-to-value ratio ϕ lowers the equity $(1-\phi)P_t h_t^O$ a buyer needs and allows an earlier purchase (Figure 5b), but it leaves the cost of a child's space unchanged: by (3.4), that space costs about $r_t h_P$ every period, whether the household rents or owns.[8] If births respond to ϕ, the borrowing limit stands between some households and a child. For a change in ϕ in period t, the envelope theorem, (3.8) and (3.5) give[9]

$$\begin{aligned} \frac{\partial \pi_{t,a}^F}{\partial \phi} &= \frac{\pi_{t,a}^F\left(1-\pi_{t,a}^F\right)}{\kappa}\,\pi_a\left[\frac{\partial V_t^H(n_t+1)}{\partial \phi} - \frac{\partial V_t^H(n_t)}{\partial \phi}\right], \\ \frac{\partial V_t^H(n)}{\partial \phi} &= \sum_{h^O \in \mathcal{H}} \pi_t^H(h^O \mid n)\,\zeta_t(h^O, n)\,P_t h^O, \end{aligned} \tag{3.9}$$

where $\pi_t^H(h^O \mid n)$ is the probability of owning h^O and $\zeta_t(h^O, n)$ the multiplier in that option. The second line is the value of credit to a household with n children at home: a higher ϕ lets an owner of h^O borrow $P_t h^O$ more, each unit is worth $\zeta_t(h^O, n)$, and the sum weights each home by the probability of owning it. Credit is worth nothing to a household that rents or whose limit is slack and, at given multipliers, worth more when homes are expensive. The first line says that credit raises the probability of trying only through the difference between its value with the additional

[7] Appendix C derives the Euler equation for the bond and the conditions behind the results below.

[8] Apart from the selling cost, an owner that holds a unit from t to $t+1$ pays $(1+\tau^p)P_t$ in period t and holds $(1-\delta_H)P_{t+1}$ in period $t+1$, so by (3.4) its cost, in period-t units, equals the rent r_t.

[9] Here $\kappa = \kappa_1$ before the first child and κ_C after. A permanent change in ϕ adds the corresponding terms for all future periods, discounted and weighted by survival.

child and without it. The difference is positive when the child moves the household toward owning larger homes, where the limit binds, or tightens the limit for a given home. It is close to zero for a household that rents in either case, and negative when credit is worth more without the child, as when it mainly helps households without children to buy. The factor in front is largest for households close to indifferent about trying and falls with age through π_a. Whether this margin matters for fertility is a quantitative question, which Section 6.3 assesses.

Earnings risk. A child stays at home for $1/\lambda$ periods on average, while earnings are risky and persistent. Consider a renter at the borrowing limit, $b_t = b_{t+1} = 0$, that rents below the cap. Maximizing (3.1) subject to $c_t + r_t h_t^R = y_t + T_t$ gives the flow utility

$$u(c_t, h_t^R, n_t) = \frac{1}{1-\sigma}\left[\frac{\alpha_0^{\alpha_0}(1-\alpha_0)^{1-\alpha_0}}{r_t^{1-\alpha_0}}\,\frac{y_t + T_t - r_t\bar{h}(n_t)}{e(n_t)}\right]^{1-\sigma} + \psi_t\, v(n_t). \tag{3.10}$$

The child's space is a committed expenditure, paid out of income before the household divides the rest between consumption and additional space. Let $\Delta_t(y_t)$ be the utility cost of a first child: the first term of (3.10) with $n_t = 0$ minus the same term with $n_t = 1$. For $\sigma \geq 1$, Δ_t is decreasing and convex in earnings, and more convex when $r_t h_P$ is larger.[10] By Jensen's inequality, earnings risk raises the expected cost of a child above its cost at expected earnings, and by more when the child's space is more expensive (Figure 6). Savings soften this effect.

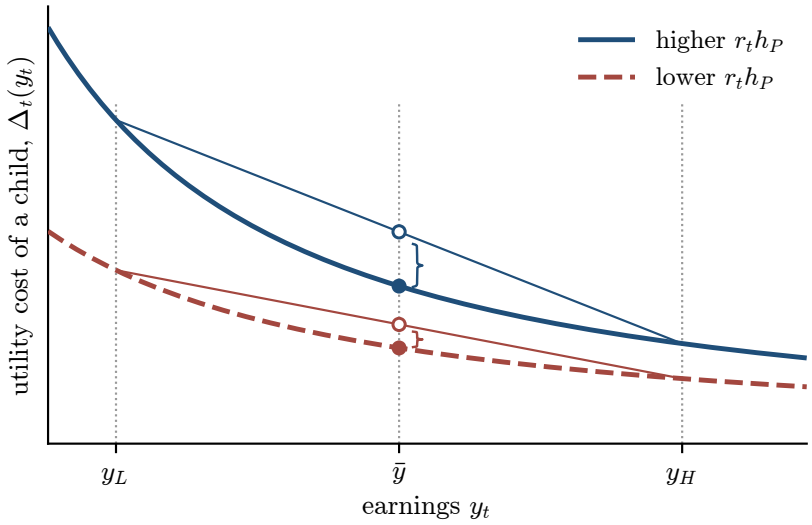


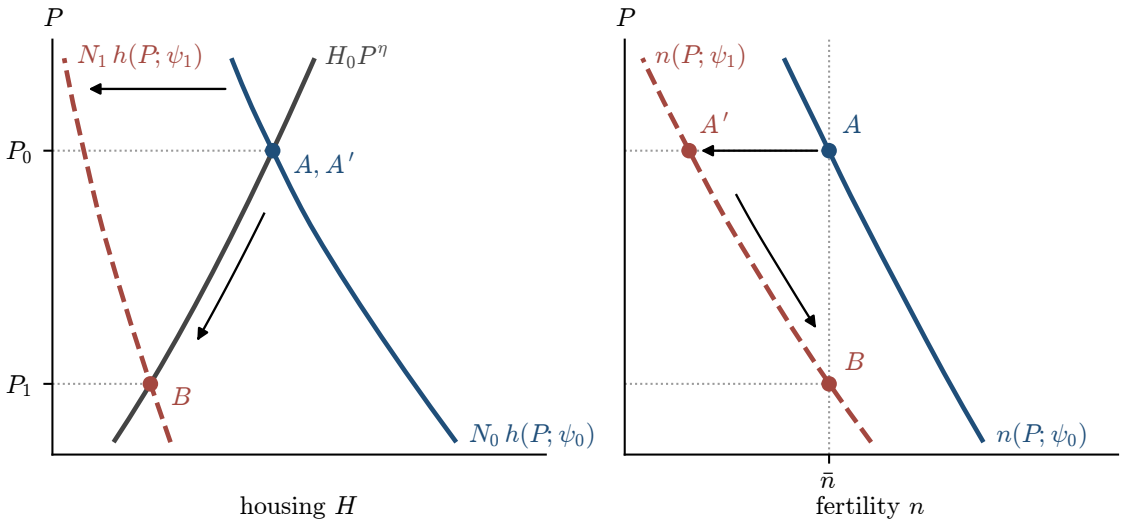
Figure 6: Earnings risk and the utility cost of a child

Notes: The figure is computed from (3.10) at the parameters of Section 4 (Appendix C). The curves plot Δ_t for a renter at the borrowing limit, at the rent of the initial stationary equilibrium (solid) and at the long-run rent of Figure 7 (dashed). Earnings are y_L or y_H with equal probability. Filled circles mark the cost at expected earnings $\bar{y}$ and open circles the expected cost; the braces mark the additional cost due to risk.

[10] Let $\tilde{y}_t = y_t + T_t$, $k = r_t h_P$ and $\Theta_t = \alpha_0^{\alpha_0}(1-\alpha_0)^{1-\alpha_0} r_t^{\alpha_0 - 1}$. Then $\Delta_t''(\tilde{y}_t) = \sigma\Theta_t^{1-\sigma}\left[e(1)^{\sigma-1}(\tilde{y}_t - k)^{-\sigma-1} - \tilde{y}_t^{-\sigma-1}\right]$, which is positive for $\sigma \geq 1$, since $e(1) \geq 1$ and $k > 0$, and rises with k.

Households and the housing market. Figure 7 shows how the equilibrium responds to a fall in the preference for children, from ψ_0 to ψ_1. Panel (a) shows the housing market. Supply (3.3) rises with the price, and demand, $N\,h(P;\psi)$, is the number of households N times the housing each occupies at the price P. Panel (b) shows the fertility $n(P;\psi)$ that households choose at each price.[11] At A, the market clears at P_0 with N_0 households, and fertility is at the replacement rate $\bar{n}$. When the preference falls, the population is inherited and the price stays at about P_0, so fertility falls along $n(P;\psi_1)$, from A to A'. Over the following generations, smaller cohorts reduce the number of households, demand shifts left, and the price falls along the supply curve. The lower price raises fertility along $n(P;\psi_1)$, until fertility is back at replacement at B, with fewer households and cheaper housing. The quantitative transition of Section 4.6 follows these steps.

In a stationary equilibrium, the number of new households equals the number that die, so fertility is at the replacement rate. Since the household problem, the pension and the rebate per household do not depend on the number of households, the condition $n(P;\psi) = \bar{n}$ in panel (b) determines the long-run price, and supply and demand in panel (a) then determine the number of households. For a given initial equilibrium, a more elastic supply leaves the long-run price unchanged and implies a larger fall in the number of households.



(a) Housing market (b) Fertility

Figure 7: The housing market and fertility

Notes: The figure is computed from stationary solves of the model at given prices, at the parameters of Section 4 (Appendix C). Panel (a) shows housing supply (3.3) and demand with N_0 households under ψ_0 and N_1 households under ψ_1. Panel (b) shows the fertility households choose at each price; the dotted line marks replacement. A is the initial stationary equilibrium, A' the outcome at the inherited population after the fall in ψ, and B the new stationary equilibrium.

[11] Housing and fertility respond to the rent. In a stationary equilibrium, the rent is proportional to the price by (3.4), so both can be read against the price. Along a transition, the rent also reflects the expected change in the price, by (3.4).

4 Quantification

I treat the United States as an economy in transition to a smaller population. I calibrate a stationary economy for 2007, before the decline in fertility, to fertility, housing and wealth moments. Holding these parameters fixed, I estimate a path of the preference for children ψ_t that matches period fertility through 2023, and project the economy forward from 2023. The policy experiments of Section 6 all start from this 2023 economy.

4.1 Data

Table 1 lists every moment with its source. Fertility moments come from the June Fertility Supplement of the CPS, pooling 2004 and 2006: childlessness and the share of mothers with exactly one child among women aged 40 to 44, and children ever born by age 25, capped at three. The mean age at first birth comes from the NCHS natality files for 2003–2006. Mean rooms come from the 2007 AHS, for heads aged 18 to 85. Ownership among heads aged 30 to 55, and the gap in ownership between recent parents, whose oldest child is under four, and heads without children, come from the 2005–2006 ACS.[12] The room response to the first birth is the PSID event-study estimate of Section 2 on the common support of the panel and the model, 1.44 rooms, and I apply the same estimator to simulated households.[13] Wealth is aggregate net worth over aggregate annual gross labor earnings in the 2005 and 2007 PSID,[14] and the bequest flow is the annual flow of estates to children relative to aggregate net worth in the 2007 SCF.

4.2 From annual data to four-year periods

A model period lasts four years. The unit of account is mean annual gross labor earnings of working-age households, \$82,882 in the 2005 and 2007 PSID (2022 dollars). Stocks, such as wealth and house prices, are in annual units; flows, such as income, consumption and rent, accrue over the four years. Interest, discounting and depreciation compound over the period, so $R_b = 1.02^4$ and $\beta = \beta_{\text{ann}}^4$, and the property tax is four times its annual rate. Children leave home after 18 years on average, so each child leaves with probability $\lambda = 2/9$ per period, and π_a is the probability of a conception within four years at age a. The earnings process is estimated directly on non-overlapping four-year means of household earnings in the 1984–1997 PSID, the last years in which the survey is annual.[15]

[12] Both ownership samples exclude mobile homes, boats and vans (ACS structure codes 3 to 10), which have no counterpart in the model.

[13] Model first births are dated from the birth event and model housing is mapped to the PSID interview calendar, so model and data compare the change in rooms over the same event-time windows.

[14] Net worth of heads aged 18 to 85, excluding business and farm equity and other real estate; gross labor earnings of heads aged 18 to 65.

[15] After removing age and year effects, ρ_z comes from the first two autocovariances of log four-year earnings. A data statistic for an age window weights each four-year model cell by the share of the cell inside the window; for the mean age at first birth, model births are dated at the midpoint of their cell and the natality data are grouped into the same cells.

4.3 Parameters set outside the estimation

I set the parameters in the left half of Panel B of Table 1 outside the estimation. Some are measured directly in the data; the others follow the literature.

Demographics. Households enter at 18, retire at 66 and live at most through 85. Survival is certain before retirement and, after it, follows the 2023 period life table of the Social Security Administration. A child leaves home with probability $\lambda = 2/9$ per period, so children stay at home for 18 years on average. Births are possible through age 45. The success of an attempt falls with age following the exponential schedule of Sommer (2016), from 0.98 at 18 to 0.90 at 30, 0.80 at 35 and 0.62 at 40.[16]

Preferences. Risk aversion is $\sigma = 2$. The equivalence scale is that of Scholz et al. (2006), with $\omega = \nu = 0.7$: a couple with two children needs 1.45 times the consumption, and the housing services above its requirement, of a childless couple to reach the same utility. The consumption weight $\alpha_0 = 0.733$ is measured directly. A childless renter has no space requirement and no owner multiplier, so its utility is Cobb–Douglas in consumption and rented space, and it spends a share $1 - \alpha_0$ of its budget on rent. In the 2019–2023 CEX, childless renters spend 26.7% of their consumption expenditure on housing.

Housing finance and taxes. The annual real interest rate is 2%, and selling a home costs 6% of its value, as in Greaney et al. (2025). Buyers can finance at most $\phi = 0.80$ of the price, the conventional limit above which a mortgage requires private insurance, and there is no limit on payments relative to income. I measure depreciation and the property tax at the national level, with the method of Greaney et al. (2025). Depreciation, 1.4% a year, is the 2007 depreciation of owner-occupied residential structures in the BEA fixed-asset accounts, divided by the value of the housing stock, land included, in the Financial Accounts of the Federal Reserve. The property tax, 1.06% of value, is the ratio of aggregate property taxes to the aggregate value of owner-occupied homes in the 2007–2011 ACS. Retirees receive a common pension equal to 23% of mean earnings, the ratio in the CPS ASEC, and the payroll tax that finances it is 8%.

Earnings. Earnings follow the four-year process of Section 4.2. In the PSID, the variance of log four-year earnings, net of age and year effects, is 0.51, and its first autocovariance is 0.37. These give $\rho_z = 0.735$ and $\sigma_z = 0.484$, which I approximate with nine states. The second autocovariance, which the estimation does not use, is 0.29 in the data and 0.27 in the process.

[16] A birth into the state of three or more children counts as 3.60 children in completed fertility.

Housing. Owners choose among homes of 2, 4, 6, 8 and 10 rooms, and renters any size up to $\bar{h}^R = 6$ rooms. The cap reflects the rental stock: in the 2023 AHS, 43% of owner-occupied homes have seven or more rooms, against 7.5% of rented homes (Appendix A). The supply elasticity is $\eta = 0.63$, from the estimates of floor-space supply of Baum-Snow and Han (2024); rooms are closer to floor space than to housing units. A higher elasticity, $\eta = 1.5$, as in Kaplan et al. (2020), is the alternative.

Entry wealth and bequests. New households draw their initial wealth from the PSID. I take the net worth of childless renter heads aged 18 to 24 in 1984–2019, set negative values to zero, and scale it to model units by the ratio of mean model earnings at entry to mean income in the sample. Entrants receive these wealth levels by rank of income, so that entrants with higher earnings also bring more wealth. The bequest shifter is $\theta_1 = 0.008$, as in Greaney et al. (2025).

4.4 Internally estimated parameters

I estimate the remaining ten parameters, in the right half of Panel B of Table 1, to match the ten moments in Panel A. They minimize

$$\sum_{k=1}^{10} w_k \big[m_k(\theta) - \hat{m}_k \big]^2, \tag{4.1}$$

where $\hat{m}_k$ is a data moment, $m_k(\theta)$ its counterpart in the stationary equilibrium at parameters θ, and w_k the inverse square of a tolerance for the moment. Completed fertility is normalized to replacement, 2.1: the house price solves this condition, and H_0 is the supply scale at which that price clears the housing market with one unit of households.

Even though the parameters are jointly identified, some moments are more informative about some parameters than others. Below I discuss which variation in the data pins down each parameter, and through which mechanism of the model.[17]

Family size. The CPS shows how the births of a cohort are spread across women. Among women aged 40 to 44 in 2004 and 2006, 19.8% are childless, and 21.4% of mothers have exactly one child. At a given completed fertility, the first share measures how many women become mothers, and the second how many children mothers have. I choose the first-birth cost $\xi = 0.295$ and the curvature $\gamma = 0.646$ to match the two shares. The cost is paid only at the first birth, so it lowers the value of becoming a parent and leaves the value of later children unchanged: a higher ξ raises childlessness, and the same births then come from fewer mothers with larger families. A higher γ makes the benefit of an additional child decline faster, so the first child is more valuable relative to later ones:

[17] The responses quoted below are elasticities from a 1% change in one parameter, solving for the equilibrium, house price included, each time.

childlessness falls and the share of mothers with one child rises.

Timing of births. In the NCHS natality files for 2003–2006, the mean age of mothers at first birth is 26.0 years. In the model, the timing of the first child trades off the decline of π_a with age against the rise in earnings and savings that makes the space for a child affordable (Section 3.4). The scale of the first-birth taste shocks, $\kappa_1 = 0.050$, governs how closely first births follow this trade-off rather than idiosyncratic tastes, and I choose it to match the mean age at first birth. The scale for later births, $\kappa_C = 0.129$, plays the same role for second and higher births. Children ever born by age 25, 0.81 in the CPS, carries information on early births, but no parameter moves it much; I return to this miss in Section 4.5.

Preference for children. Completed fertility is held at replacement, so the preference for children $\psi_0 = 0.177$ cannot be read from the number of births. A higher ψ_0 raises the demand for children, and the house price rises until births return to 2.1: a 1% higher ψ_0 raises the price by 3.0% and lowers mean rooms by 1.9%, and barely moves childlessness or the share of mothers with one child. I therefore read ψ_0 from the level of housing demand at replacement: mean rooms, 5.73 in the 2007 AHS, and ownership. Intuitively, the more households value children, the higher the price of space at which they still have 2.1 children, and the less space they occupy.

Housing. I choose the owner multiplier $\chi = 1.112$ to match ownership among heads aged 30 to 55 in the 2005–2006 ACS, 67.6%. An owner obtains χ times the services of the same home, so a higher χ raises the value of owning relative to renting: a 1% higher χ raises ownership by 3.2%.

The tenure taste scale $\kappa_T = 0.013$ and the space requirement $h_P = 0.89$ rooms are disciplined by how housing differs between parents and other households. In the same ACS sample, heads whose oldest child is under four own at a rate 12.8 percentage points higher than heads with no children at home. In the model, children at home raise the space a household needs by h_P, and homes larger than $\bar{h}^R$ rooms are only for sale, so parents move into ownership. A higher h_P widens this gap, and so does a smaller κ_T, which makes tenure follow these needs more closely. Mean rooms separate the two: they rise with h_P and barely move with κ_T.

The room response to the first birth adds the change in housing within households. Two to three years before the first birth, future parents in the PSID live in 4.71 rooms; three to four years after it, they occupy 1.44 more rooms, relative to the change among adults who never have a child (Section 2.2). I apply the same estimator to simulated households. In the model, this response reflects both the space a child requires and the rooms households add as their earnings and savings grow with age, so it is not tied to h_P alone. Across the estimated parameters, it moves closely with the house price: a 1% higher ψ_0 raises the price by 3.0% and the response by 1.8%.

Table 1: Parameterization

A. Moments used in the estimation

Moment	Source	Data	Model
Completed fertility (normalization)	Replacement	2.10	2.10
Childless, women 40–44 (%)	CPS 2004, 2006	19.8	20.1
Exactly one child among mothers, 40–44 (%)	CPS 2004, 2006	21.4	21.7
Mean age at first birth (years)	NCHS 2003–2006	25.98	25.97
Children ever born by age 25	CPS 2004, 2006	0.810	0.556
Mean occupied rooms	AHS 2007	5.73	5.75
Ownership, heads 30–55 (%)	ACS 2005–2006	67.6	66.9
Recent-parent ownership gap	ACS 2005–2006	0.128	0.125
First-birth room response	PSID event study	1.440	1.475
Wealth / annual gross earnings	PSID 2005, 2007	4.46	4.35
Annual bequests / wealth (%)	SCF 2007	0.73	0.73

B. Parameter values

Assigned			Estimated		
σ	2	risk aversion	β_{ann}	0.936	annual discount factor
(ω, ν)	0.7, 0.7	equivalence scale	ψ_0	0.177	preference for children
α_0	0.733	consumption weight	γ	0.646	curvature, benefit of children
λ	2/9	child departure, per period	ξ	0.295	first-birth utility cost
π_a	0.98–0.50	success of a birth attempt	κ_1	0.050	first-birth taste scale
ϕ	0.80	financed share	κ_C	0.129	later-birth taste scale
μ	0.06	selling cost	h_P	0.89	space requirement (rooms)
δ_H	0.014	depreciation, annual	χ	1.112	owner service multiplier
τ^p	0.0106	property tax, annual	κ_T	0.013	tenure taste scale
r	0.02	real interest rate, annual	θ_0	0.244	bequest-motive scale
τ^w	0.08	payroll tax	H_0	6.776	housing supply scale
η	0.63	housing-supply elasticity			
$\bar{h}^R$	6	rental size cap (rooms)			
θ_1	0.008	bequest shifter			
(ρ_z, σ_z)	0.735, 0.484	earnings, per four years			

Notes: This table reports the moments used in the estimation and the parameter values. Samples and definitions in Section 4.1. The model's room response is measured with the event-study design of the data on simulated households. Owners choose among homes of 2, 4, 6, 8 and 10 rooms. The 2007 house price is 0.771; the weighted loss (4.1) is 7.42, of which children ever born by 25 contribute 6.41.

Wealth and bequests. I choose the discount factor $\beta = 0.936$ (annual) to match aggregate net worth of 4.46 times annual gross labor earnings in the 2005 and 2007 PSID. More patient households save more, so wealth rises with β. Since savings decide when a household can pay for the space a child requires, β also moves the timing of first births and the room response. The bequest scale $\theta_0 = 0.244$ governs how much wealth households keep for their heirs in old age. I choose it to match the annual flow of estates to children in the 2007 SCF, 0.73% of aggregate net worth. This flow is the only target that responds to θ_0.

Table 2: Moments not targeted, 2007

Moment	Source	Data	Model
Mothers by age 25 (%)	CPS 2004, 2006	45.7	45.5
First births at age 30+ (%)	NCHS 2003–2006	24.9	24.2
Rooms: 3+ versus 1–2 resident children	ACS 2005–2006	0.385	0.574
First-birth ownership response	PSID event study	0.261	0.141
Wealth/income p90 / median, ages 76–84	PSID	3.52	3.27

Notes: This table reports moments that do not enter the loss (4.1). The ownership response is the event-study coefficient of Section 2 on the model's observation support, measured on simulated households with the same design.

4.5 Testing the model

I next ask how well the model accounts for features of the data that the estimation does not target. Table 2 reports untargeted moments, and Figure 8 compares life-cycle profiles. I focus on the timing of births and on how households adjust housing and tenure around a birth, the two margins through which housing policy acts in Section 6.

Fertility. The model matches the targets closely except children ever born by age 25, 0.56 against 0.81 (Panel A of Table 1). The share of women who are mothers by 25 is matched, 45.5% against 45.7%, so the miss is in the number of children of young mothers: 1.2 at 25 in the model, against 1.8 in the data. A household can have at most one birth in a four-year period, so in the model no woman has three children by 25 and few have two; no parameter can change this. Later first births are close to the data: 24.2% of first births occur at 30 or later, against 24.9%. The right panel of Figure 8 shows the whole distribution. The model places fewer first births between 26 and 33, 28.5% against 33.6%, and more at 18 to 21 and after 38.

Housing around the first birth. The model reproduces the size of the room response after the first birth, which it targets, but not its timing. In the data, parents occupy the same number of rooms in the years before the reference period, add 0.62 rooms in the window of the birth and 1.44 three to four years after it, and keep adding rooms, to 1.8 more eleven or more years after the birth. In the model, households add 0.78 rooms in the window of the birth and 1.48 three to four years after it. But they already add rooms in the years before the birth: eight or more years before it they occupy 0.65 fewer rooms than in the reference period, and four to five years before it 0.20 fewer. Part of the model's response therefore reflects the growth of housing with age and income among future parents, rather than the birth itself. After the birth, model households stop adding rooms, and eleven or more years after it the response falls to 0.65 rooms.

Tenure responds less than in the data. Ownership rises around the first birth by 14 percentage points in the model, against 26 in the data, consistent with the higher ownership of young households in the model (below). Larger families also add more space than in the data: households with three or

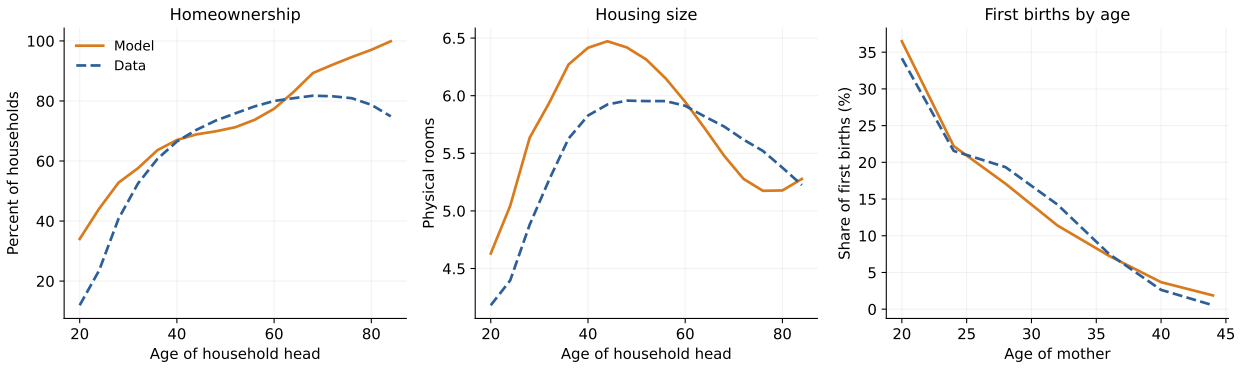


Figure 8: Life-cycle profiles in 2007, model and data

Notes: Homeownership and rooms by age of the household head, and the distribution of first births by age of the mother. Model: 2007 steady state. Data: ownership and rooms (at most nine) from the 2005–2006 ACS; first births by four-year age group from the NCHS natality files for 2005–2007.

more children occupy 0.57 more rooms than those with one or two, against 0.39.

Life-cycle profiles. The left and middle panels of Figure 8 compare ownership and rooms by age of the head with the 2005–2006 ACS. The model reproduces the rise of ownership with age, and between 34 and 49 the two profiles are within four percentage points. Young households own too often: 34% at 18–21 and 44% at 22–25, against 12% and 24% in the data. At old ages, ownership keeps rising in the model, to nearly all households after 80, while in the data it falls from 82% at 66–69 to 75% at 82–85. The model has no health or long-term-care risk that would lead older owners to sell. The model also reproduces the hump in housing size, but its profile is steeper: rooms rise from 4.6 at 18–21 to a peak of 6.5 at 42–45, against 4.2 and 6.0 in the data, and then fall faster. The ACS records at most nine rooms, and its profile lies below the AHS mean that the model targets, so the comparison is informative about shape rather than level.

Wealth. The model accounts for the dispersion of wealth among the old: the ratio of the 90th percentile to the median of wealth relative to income at ages 76 to 84 is 3.27, against 3.52 in the PSID.

4.6 The fertility decline, 2007–2023

Figure 9 shows the decline to be explained. The period fertility rate of the United States peaked at about 2.1 in 2007 and fell to 1.62 by 2023. Completed fertility of women aged 40 to 44, which reflects births over the previous two decades, was 1.92 in 2024. The constant-rate scenario in the figure makes completed fertility converge to 1.62 within about two decades as the cohorts that had their children before 2007 age out. This continuation illustrates the distinction between a current flow of births and the lifetime fertility of older cohorts; it is not an estimate of future age-specific birth rates.

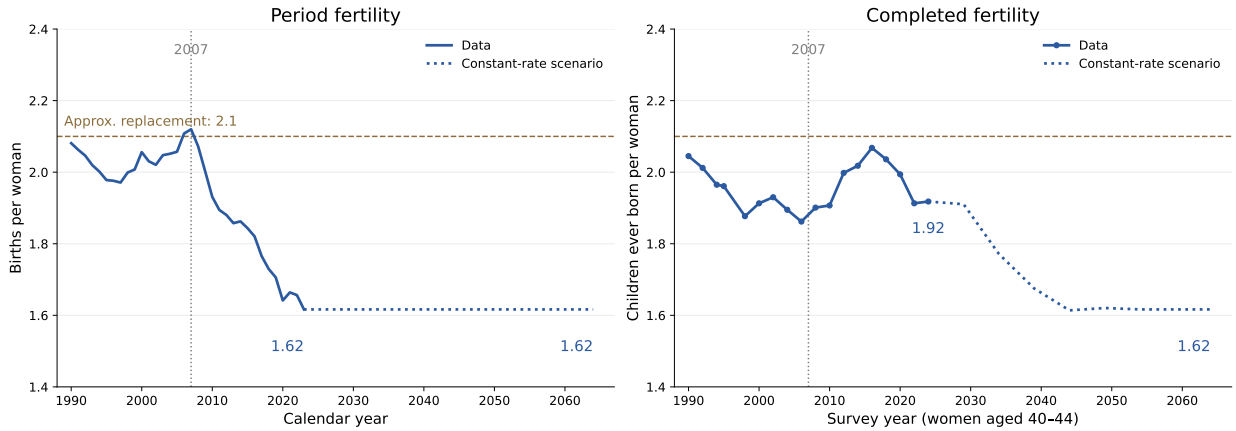


Figure 9: Period and completed fertility in the United States

Notes: Left: period total fertility rate, World Development Indicators. Right: children ever born to women aged 40–44, CPS Fertility Supplement. Dotted lines: a constant-rate scenario that holds period fertility at its 2023 level; the completed-fertility continuation is rescaled to end at that rate.

Starting from the 2007 steady state, I hold all estimated parameters fixed and let the preference for children decline permanently from its 2007 level ψ_0 toward a lower level ψ',

$$\psi_t = \psi' + (\psi_0 - \psi')\,\rho^{\tau}, \tag{4.2}$$

where τ counts four-year periods since 2007. Households learn the whole path in 2007 and have perfect foresight about it and about prices. The house price, the pension and the rebate clear their markets at every date (Appendix E). I set $\rho = 0.35$ and choose $\psi' = 0.147$ to match period fertility in 2019–2022, 1.646. The model then gives 1.73 in 2011–2014, against 1.86 in the data (Figure 10). The decline in ψ_t summarizes changes in the incentives to have children that the model does not otherwise capture; its role is to match the fall in period fertility while prices, family composition and household entry adjust. After 2023, period fertility is 1.64 in 2023–2026 and rises slowly, to 1.80 by 2063.

[Problem: The transition in this subsection is fitted with the slowly adjusting stock of Appendix D.1. The counterfactuals of Section 6 start from the 2023 state of the same fit under free entry, (3.3), in which period fertility is 1.63 in 2023 and the price is 3.6% below its 2007 level. Which supply rule is the main case is under decision.]

Figure 11 shows the whole transition. Period fertility approaches replacement slowly: it is 2.09 by 2399, when the number of households is 49%, total housing 67% and the house price 52% of their 2007 levels. The housing stock adjusts slowly, so the price falls ahead of the stock: in 2023 the price is 9.0% below its 2007 level and the stock 1.1% below; in 2063 they are 29.2% and 6.6% below. The rent falls less than the price early on. Along a transition, rent equals the user cost net of the expected capital gain, so a price that is expected to keep falling holds up the rent: in 2023 the rent per room is only 0.3% below its 2007 level.

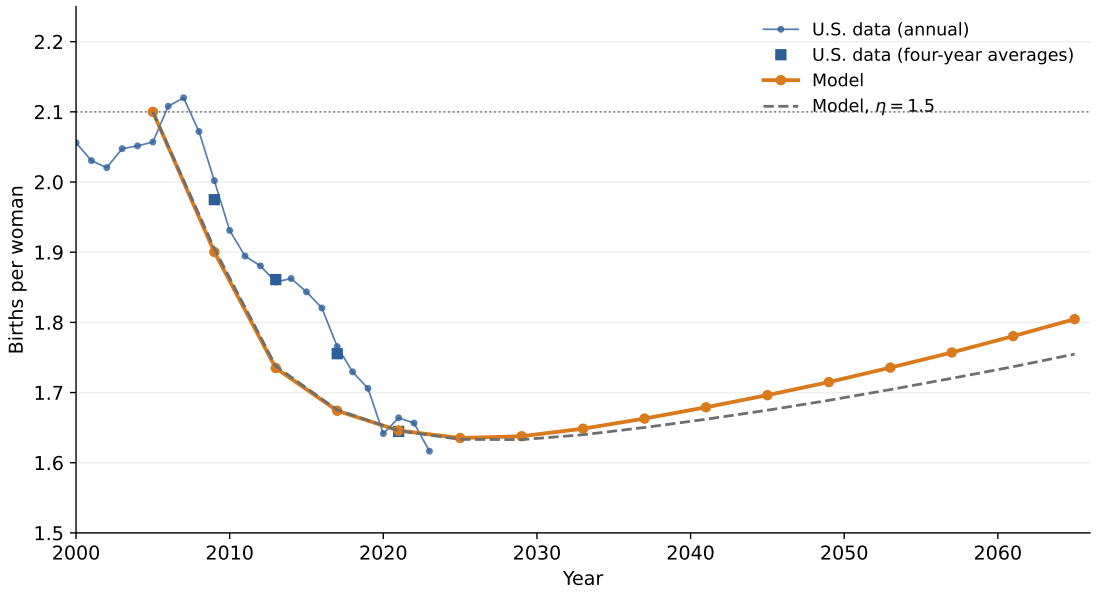


Figure 10: Period fertility, model and data, 2007–2063

Notes: Model: births per woman in each four-year period of the transition from the 2007 steady state, with $\eta = 0.63$; dashed, the same transition with $\eta = 1.5$. Data: annual period fertility and its four-year averages.

4.7 The 2023 economy

The 2023 economy is the starting point of every counterfactual, so I compare it with the data on the moments used in the calibration (Table 3). None of these moments was targeted in the transition, which matches only the period fertility rate. Mean rooms match the data, 5.58. Ownership among heads aged 30 to 55 is 73.7% against 58.7%: in the data ownership fell after 2007, while in the model, where housing became cheaper, it rose. The model misses the postponement of first births: the mean age at first birth is 26.9 against 28.1, and 29% of first births are at 30 or later against 38%. Households with three or more children occupy 0.54 more rooms than those with one or two, against 0.36 in the data, and wealth is 4.32 times annual earnings against 7.05.

The fertility moments reveal the limitations of explaining the decline with changes in the preference for children alone. In the data, the decline in period fertility since 2007 came with almost unchanged childlessness among women aged 40 to 44 (19.8% in 2004–2006, 18.8% in 2024) and with later first births: the mean age at first birth rose from 26.0 to 28.1 and the share of first births at 30 or later from 25 to 38%. In the model, a lower ψ_t works on every cohort alive, so 29% of women aged 40 to 44 in 2023 are childless and their completed fertility is 1.69, against 1.92 in the data. The data show a prominent role for later first births among younger cohorts; the model instead reduces fertility across the cohorts exposed to the common preference change. This discrepancy leaves the distribution of preference changes across cohorts unresolved.

With only preference shocks, falling fertility lowers the demand for family space, so the model has to deliver a falling price: 9.0% below 2007 in 2023. In the data, the real house price rose by 15.6%

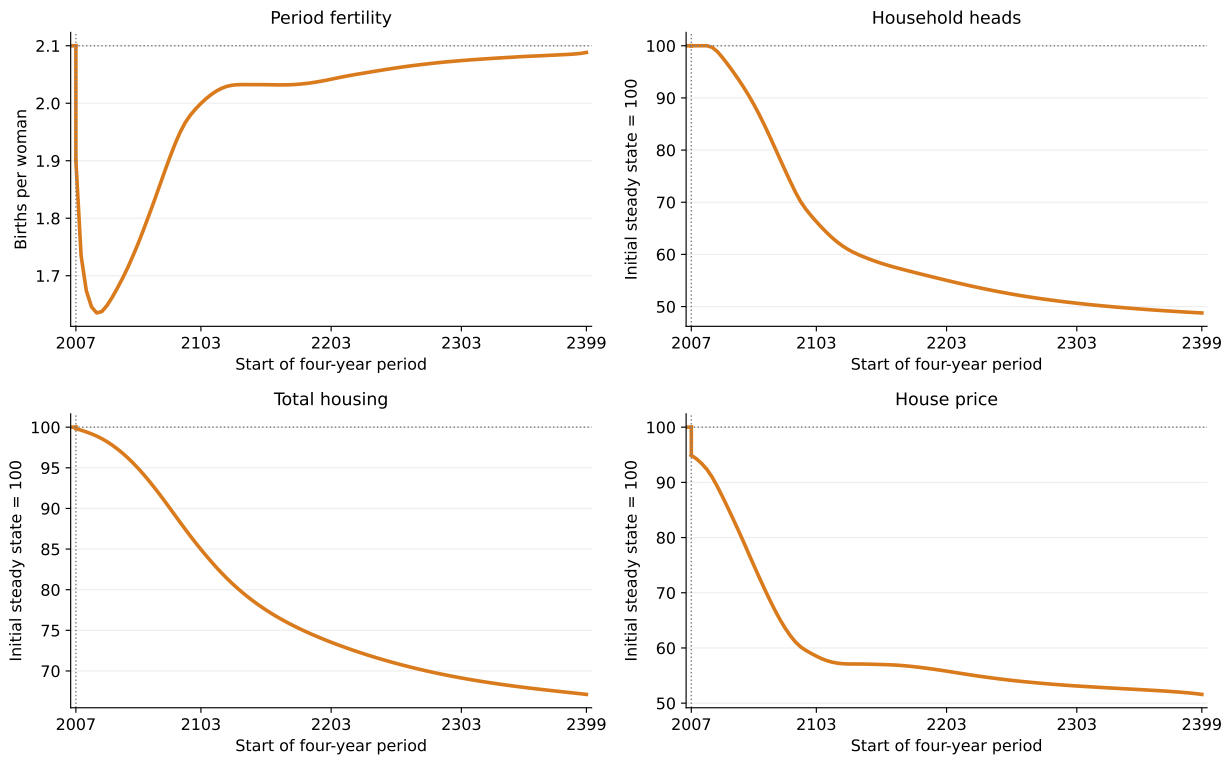


Figure 11: The full transition, 2007–2399

Notes: Period fertility, number of household heads, total housing and the house price along the transition with $\eta = 0.63$, with the initial steady state normalized to 100.

between 2007 and 2023, and real rents by 15 to 19%. Part of the decline in fertility since 2007 therefore came with a rising cost of space that the estimation assigns to preferences. The transition says what housing would have done in response to the fertility decline, not that housing caused it.

Figure 12 compares life-cycle profiles in 2023. The model reproduces the rise of ownership with age and the hump in housing size, but its households buy too early: 45% of heads aged 18 to 21 and 57% of those aged 22 to 25 own their home in the model, against 11% and 15% in the ACS. This miss matters for the credit experiments below, because it places too many young households near the margin of buying. First births are also too early: 29% of first births in the model are to women aged 18 to 21, against 17% in the 2021–2023 natality data, and 14% are at 30 to 33, against 22%.

5 Household Choices in the Calibrated Model

This section describes how households in the calibrated model respond to parenthood, and how earnings risk shapes the decision to have one. First, I show how having children changes housing, consumption and tenure. The child's space trades off with consumption, so the cost weighs most on households with little wealth or low earnings. Second, I illustrate how lowering the standard deviation of earnings shocks by about a quarter, holding the house price at its 2007 level, raises

Table 3: The 2023 economy: data and model

Moment	Data	Model
Completed fertility, ages 40–44	1.92	1.69
Childless, ages 40–44 (%)	18.8	28.8
Exactly one child among mothers, 40–44 (%)	22.2	22.3
Mean age at first birth (years)	28.09	26.89
Mothers by age 25 (%)	27.4	32.3
First births at age 30+ (%)	37.7	29.0
Mean occupied rooms (capped at 9)	5.58	5.58
Ownership, heads 30–55 (%)	58.7	73.7
First-birth room response‡	1.476	0.55
Rooms: 3+ versus 1–2 resident children	0.355	0.538
Wealth / annual gross earnings (PSID 2021/23)	7.05	4.32
Annual bequests / wealth (%)	0.88	0.83
Wealth/income p90 / median, ages 76–84	3.45	3.44

Notes: Model: the 2023 period of the transition in Section 4.6, evaluated on the full 2023 distribution. ‡ The model value compares otherwise identical households with and without a first birth in the same period, not the event study of the data. Data: the sources of Table 1 in their most recent waves; wealth over earnings from the PSID 2021 and 2023.

completed fertility by 12%.

5.1 Choices at the first birth

Figure 13 compares the choices of the same household in the period in which it has no child and in the period in which a first child is born. The household is a renter aged 26 to 29 with median earnings, and the figure varies its liquid wealth. Without savings, a first child pushes the space the household rents close to the rental cap and lowers the probability of owning from 42% to 24%. The child needs space when the household can least afford a down payment, so it rents and pays for the space out of consumption. With more than about two years of earnings in liquid wealth, the child

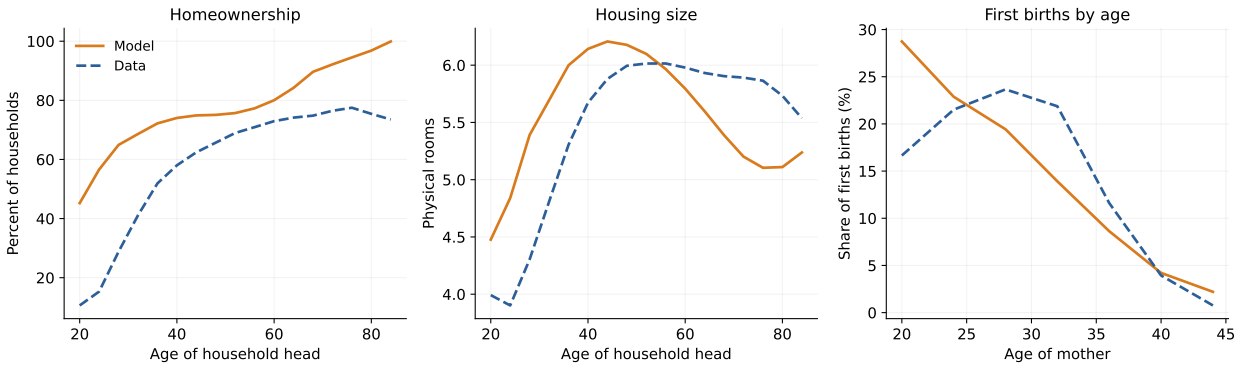


Figure 12: Life-cycle profiles in 2023, model and data

Notes: Homeownership and rooms by age of the household head, and first births by age of the mother. Model: the 2023 period of the transition in Section 4.6. Data: ACS 2023; NCHS first births 2021–2023.

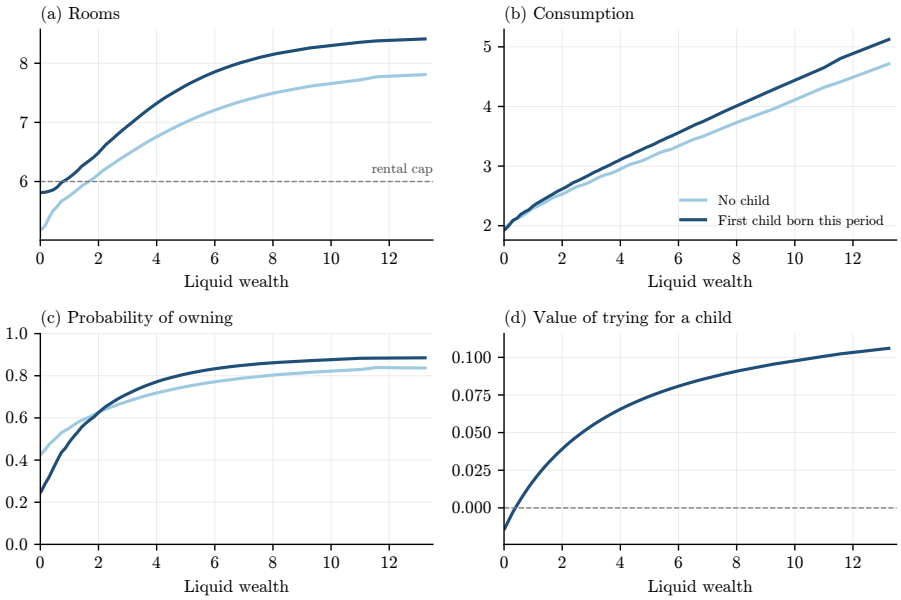


Figure 13: Choices with and without a first child

Notes: Renter aged 26–29 with median earnings ($z = 0.78$), 2007 stationary equilibrium. Liquid wealth in units of mean annual earnings. Rooms and consumption are averages over tenure choices.

instead raises the probability of owning, and the household buys a larger home.

Panel (d) shows the value of trying for a child, $V^F_{t,A} - V^F_{t,0}$. It is negative without savings, turns positive at about five months of earnings, and rises steeply at low wealth before flattening. Wealth therefore matters for the decision to have a first child mostly among households with little of it.

5.2 Choices across earnings

Figure 14 repeats the comparison across earnings states, for renters aged 26 to 29 without savings. A first child adds about two thirds of a room at earnings up to the median, and about half a room at high earnings, where most of these households buy a home even without a child. Consumption falls by 21% at the lowest state and by 2% at the median state. At high earnings it rises, because the equivalence scale raises the household's needs and these households can afford to meet them. As Section 3.4 shows, the space requirement alone costs about $r_t h_P$ in every period the child is at home, whether the household rents or owns. At the 2007 price, this is 2.9% of mean annual earnings each year. The cost of a child's space is therefore a recurring expense, and it weighs most on households with low earnings, who indeed trade it off with consumption.

5.3 Earnings risk

Section 3.4 showed that earnings risk raises the expected cost of a child, since that cost falls with earnings at a decreasing rate. To measure this force, I lower the standard deviation of the four-year

earnings shock by about a quarter, from 0.48 to 0.35, rescale the earnings states to keep mean earnings unchanged, and solve the model at the 2007 price. Lower risk raises completed fertility by 12%, from 2.10 to 2.36.[18] Figure 16 below reports the effect over a range of values of earnings risk.

Figure 15 shows where the additional births come from. In the baseline, the model matches the distribution of children ever born among women in their early forties. Lower risk shifts households out of childlessness and toward larger families: the share without children falls from 20% to 12%, and the shares with two and with three or more rise.

[18] Without the rescaling, the same change raises mean earnings by 4.9%, and completed fertility rises by 16%.

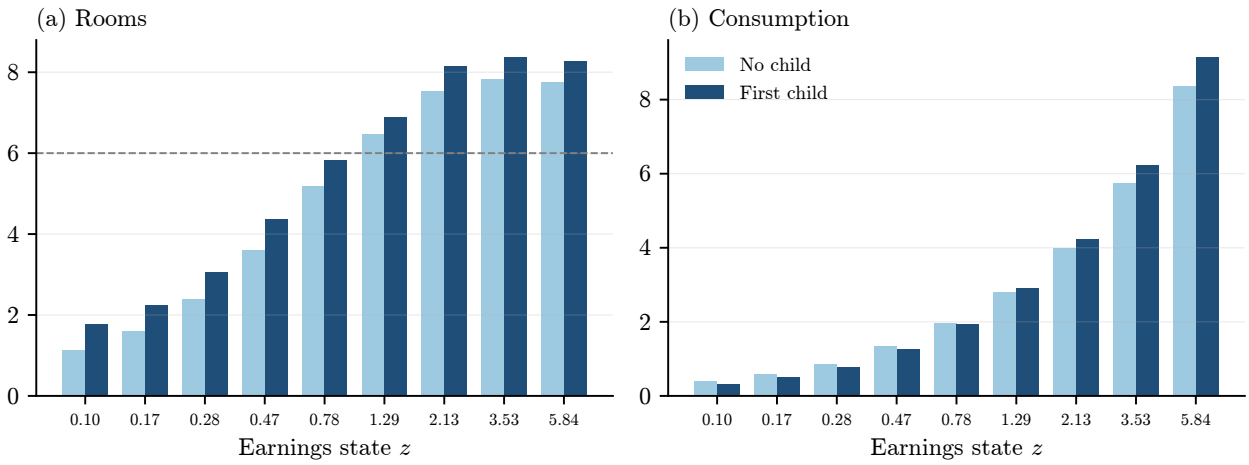


Figure 14: Choices with and without a first child, by earnings

Notes: Renters aged 26–29 without savings, 2007 stationary equilibrium, by earnings state z. Rooms and consumption are averages over tenure choices. The dashed line marks the rental cap $\bar{h}^R$.

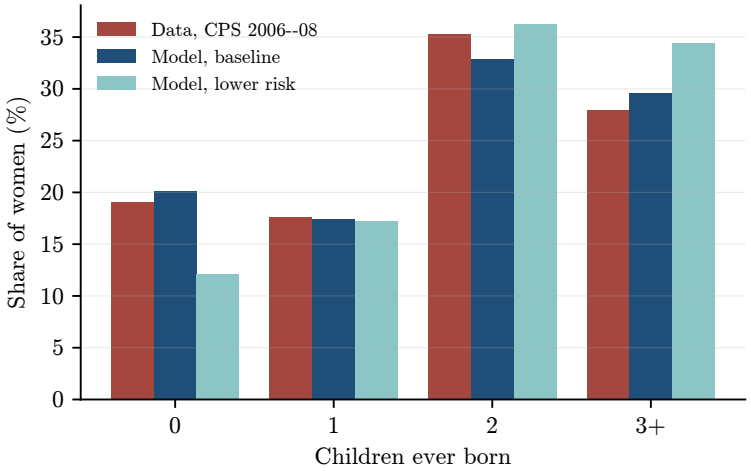


Figure 15: Earnings risk and children ever born

Notes: Shares with 0, 1, 2, and 3 or more children ever born. Data: CPS June Fertility Supplement, 2006 and 2008, women aged 40–44. Model: households at about age 42, 2007 stationary equilibrium, at the estimated earnings risk ($\sigma_z = 0.48$, baseline) and at lower risk ($\sigma_z = 0.35$, earnings states rescaled to keep mean earnings unchanged), same price.

Earnings risk weighs more when a child requires more space. Figure 16 solves the model at the 2007 price for different values of the space a child requires, h_P, and of the standard deviation of earnings shocks. Lowering risk to 0.35 raises completed fertility by 10% when a child requires half a room, and by 17% when it requires 1.6 rooms. The child's space must be paid in every period the child is at home, including after a bad earnings draw, when consumption is most valuable. The more space a child requires, the more these periods cost, and the more a household gains from lower risk.

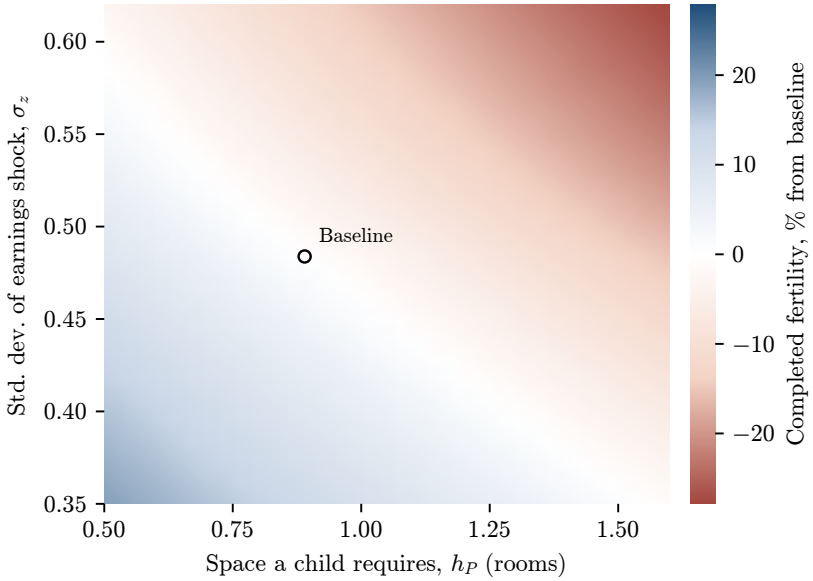


Figure 16: Earnings risk and the space a child requires

Notes: Completed fertility, % change from the baseline (marked), 2007 stationary equilibrium, same price, on a grid of 25 values of each parameter. Vertical axis: standard deviation of the four-year earnings shock, with earnings states rescaled to keep mean earnings unchanged. Horizontal axis: space a child requires, h_P, in rooms.

6 Counterfactuals

In this section, I use the calibrated model to study counterfactual changes to the 2023 economy and unpack the interaction between family space and fertility. First, I show that easing mortgage credit and expanding the housing supply raise births by the same amount on impact but have different effects over time: credit moves births earlier, while supply raises the number of children. Second, I show that credit moves births earlier because young households use it to buy the small homes they would otherwise rent: it lowers the down payment, not the cost of the space a child needs. Third, I show that credit tied to children, which does lower that cost, raises the completed fertility of the 2023 cohort by 4.9%, and I compare these changes with transfers to households.

[Problem: Numbers in red are still from the runs in the slides (instant supply, earlier calibration and accounting): the decomposition of the credit path, the split of new owners by home size, and lower earnings risk from 2023. Mortgage credit is under check (‡): it raises births by 1.35% on

impact here, against about 1.9% in the slides' runs.]

6.1 Mortgage credit and housing supply

I compare two changes that intervene in the housing market. The first eases mortgage credit: the loan-to-value limit rises from 80% to 100%, so buyers no longer need a down payment. As Section 3.4 shows, credit raises the probability of trying for a child only if it is worth more to a household with the additional child than without it (3.9): it can raise fertility when the down payment stands between a household and the home its family needs. The second expands the housing supply: the scale of the supply curve, H_0, rises by 4.35%, so that at any price the stock is 4.35% larger.[19] I choose this size so that births in 2023 rise by as much as under credit. Credit lowers the down payment needed to buy a home. Supply lowers the cost of housing, both the price and rents. With the same effect on impact, the comparison shows which of the two costs matters for fertility.

Both experiments start from the 2023 state of the transition in Section 4.6: the same households, with the same wealth, homes and children. In 2023 the change is announced unexpectedly and believed to be permanent. Households have perfect foresight of the paths of the house price, rents, the pension and the rebate, which clear their markets at every date. I compare each experiment with the path the economy follows from the same state without the change, so that the difference isolates it from the ongoing transition. By 2023 the preference for children has nearly completed its decline, and it stays close to its 2023 value along both paths.

I report births in the four-year periods starting in 2023 and 2063 and in the long run, when the economy reaches its new stationary equilibrium, and the completed fertility of the cohort aged 18 to 21 in 2023: the children ever born to these households over their lives. All are relative to the no-policy path. Table 4 summarizes the results.

Table 4: Counterfactuals from 2023

	Births			Completed fertility	House price	
	2023	2063	Long run	2023 cohort	2023	Long run
Mortgage credit‡	+1.3	+0.6	+3.0	−1.1	−0.1	+3.2
Housing supply, +4.35%	+1.4	+3.0	+4.3	+2.0	−3.9	0.0
Credit for parents	+5.1	+16.6	+36.1	+4.9	+0.3	+33.5

Notes: % differences from the no-policy path from the same 2023 state, with free entry and $\eta = 0.63$. Births: births in the four-year period starting in the year shown; long run: the new stationary equilibrium, where births per household are at replacement, so births and the number of households rise by the same percentage. Completed fertility: children ever born to the cohort aged 18–21 in 2023. Without policy, period fertility is 1.63 in 2023 and 1.73 in 2063. Housing supply: the supply curve shifted out so that births in 2023 rise as much as under mortgage credit. Credit for parents: Section 6.4. ‡ Under check.

[19] Developers enter freely, so the housing stock is on the supply curve $H_0 P_t^{\eta}$ at every date, with $\eta = 0.63$ (Baum-Snow and Han, 2024), unless that would require it to shrink faster than it depreciates. Appendix D reports the results with a more elastic supply and with a slowly adjusting stock.

6.2 Same impact, different long run

Demographers distinguish how many children are born, the *quantum*, from when they are born, the *tempo* (Bongaarts and Feeney, 1998). Credit is mostly tempo, and supply is quantum. Figure 17 shows the two paths. Both raise births by 1.3 to 1.4% in 2023. Under credit the gain is gone within four years, and births fall below the no-policy path, by 0.6% in 2035, as the births brought forward are missing later. Under supply births rise in every period, by 3.0% by 2063 and by 4.3% in the long run.

Supply raises births over time for two reasons. Cohorts that reach childbearing age after 2023 spend their whole fertile life with cheaper space, and the additional children form households of their own a generation later. Credit adds one pulse of births, and its echo fades. Supply adds a steady flow, and its echoes build up. In the long run the price returns to the level at which fertility is at replacement, and the number of households rises by the full 4.3%: the additional rooms are filled by additional families. Credit also has a small lasting effect, 3.1% more households in the long run, with a price 3.2% higher.

The path of births under credit has four phases (Figure 17, panel a). On impact, credit reaches every renter waiting to buy, at every fertile age, and some of them have their child now. From 2031 births fall below the no-policy path: households that had a child early have fewer later, and those who enter after 2023 only move their births earlier. Births then rise again. About a third of the rise is the echo, as the children born in 2023 become young parents, and two thirds comes from new entrants, who gain more from credit as housing becomes cheaper along the transition. Credit adds a little to lifetime fertility in the long run because space then becomes cheap: without policy, the price falls by 45% from 2023 to the new stationary equilibrium. At the long-run price, removing the down payment raises births 1 percentage point above replacement; at prices about 50% higher, by only 0.1 to 0.2 points. When space is expensive, the per-period cost of a family room dominates the decision, and credit, which changes only the one-time down payment, matters little. Letting renters rent homes as large as 10 rooms removes two thirds of this long-run effect.

The long run follows from one condition. In a stationary equilibrium the population is constant only if births per household are at replacement, and fertility falls with the cost of space. So the long-run price is the one at which households choose replacement fertility, and the number of households is the one that fills the housing stock at that price. A policy that raises fertility at a given price therefore ends up as a higher price and more households, not as more births per household. A supply expansion leaves the replacement price unchanged: a 10% expansion brings the price back to its no-policy level and leaves 10.0% more households. Credit for parents raises fertility at any price, so the price must rise, by 33%, until fertility is back at replacement, with 36% more households.

Table 5 follows the cohort aged 18 to 21 in 2023, which separates the number of children from their timing. Credit lowers its completed fertility by 1.1% and the share with a first birth by 1.3%, and brings first births forward by 0.08 years. The supply expansion raises completed fertility by 2.0%

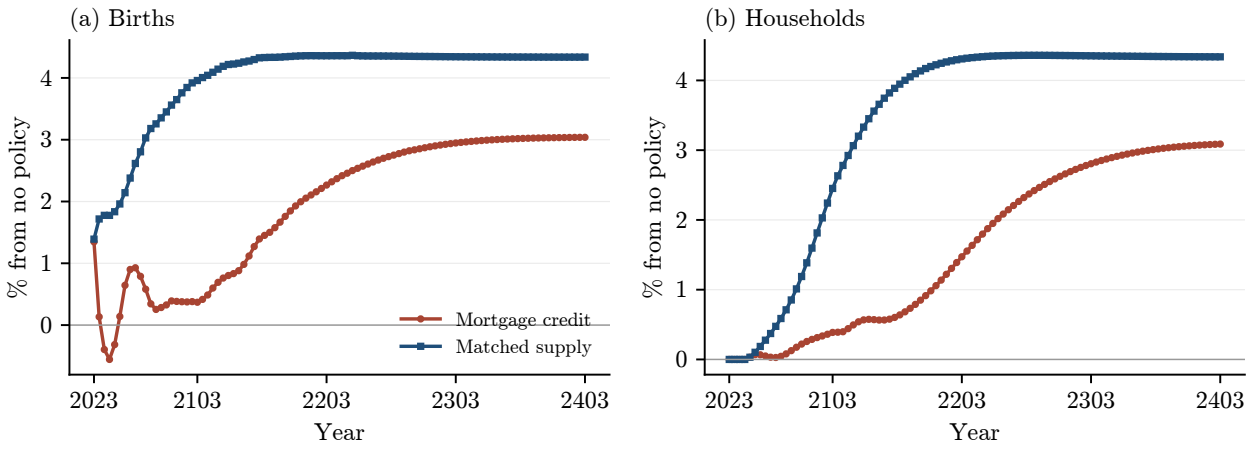


Figure 17: Mortgage credit and housing supply

Notes: % differences from the no-policy path from the same 2023 state, births in each four-year period (panel a) and the number of households (panel b), to the new stationary equilibrium. Mortgage credit: LTV 80 to 100 for all households. Housing supply: the supply curve shifted out by 4.35%.

and the share with a first birth by 1.6%. Supply also brings first births forward, by 0.10 years: a cheaper room lowers the per-period cost of the commitment, so households commit sooner as well as more often.

Table 5: The cohort aged 18–21 in 2023

	Completed fertility	Share with a first birth	Mean age at first birth
No policy	1.640	0.673	27.15
Mortgage credit‡	1.622 (−1.1%)	0.665 (−1.3%)	27.07
Housing supply	1.673 (+2.0%)	0.684 (+1.6%)	27.05
Credit for parents	1.720 (+4.9%)	0.701 (+4.1%)	26.88

Notes: Children ever born and the share with a first birth, per household of the cohort, summed over ages 18 to 45. Mean age at first birth: first-birth rates weighted by the midpoints of the four-year age cells. ‡ Under check.

6.3 Who uses the credit

Why does credit move births earlier rather than raise their number? Credit lowers the equity needed to buy a home, $(1-\phi)P_t h^O$, not the per-period cost of the space a child needs. Rent equals the user cost of owning (3.4), so a family can rent that space at about the per-period cost of owning it. By (3.9), credit raises the probability of trying for a child only if it is worth more to a household with the additional child than without it. It is worth more without the child when it mainly helps households without children to buy.

The households that use the credit confirm this. Among households aged 22 to 29 in 2023, ownership rises from 49% to 73%. Seven in eight of the new owners buy a home of 2 or 4 rooms. The share owning a home of 6 rooms or more, the size families choose, rises by only 3 percentage points. Across all households, ownership rises from 74.9% to 82.8% in 2023, and the mean size of an owned

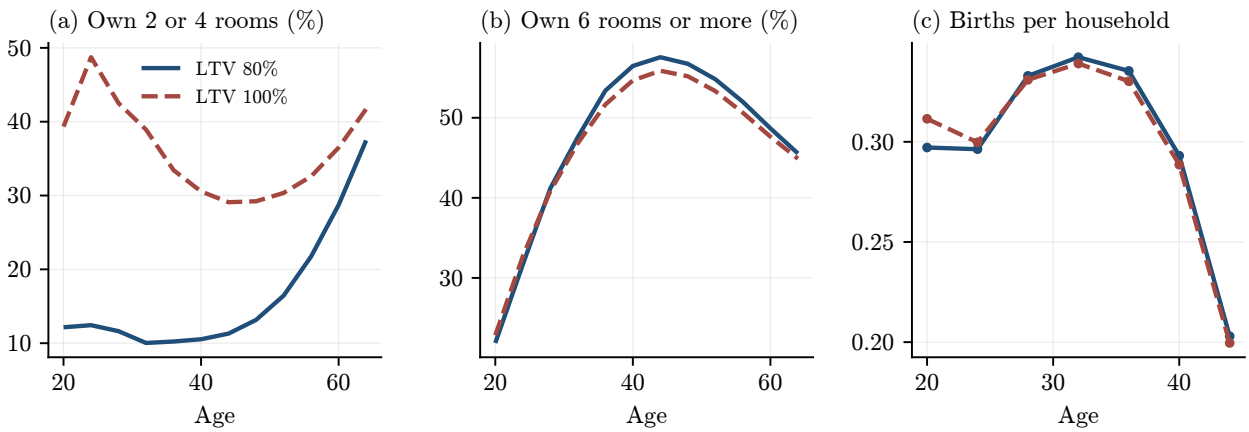


Figure 18: Mortgage credit, ownership and births

Notes: 2007 stationary equilibrium at a loan-to-value ratio of 80% and of 100%, same price. Share of households owning a home of 2 or 4 rooms (panel a) and of 6 rooms or more (panel b), and births per household in each four-year age group (panel c), by age of the head.

home falls from 6.05 to 5.73 rooms. Total demand for space falls by 0.09%, and the price with it, by 0.14%. The price barely matters for this result: at the no-policy prices, credit raises births in 2023 by 1.38%, against 1.35% in equilibrium.

Credit buys small homes early. Among households aged 22 to 25 in 2027, ownership rises from 46% to 68%, and the share owning a two-room home from 8% to 25%, while births per 100 households at those ages stay at 22.2. A small home accommodates a young family, but the households that buy one early are not those whose fertility was held back. The pattern persists along the transition. In 2059 credit raises ownership at ages 22 to 37 by 21 percentage points, and young owners live in smaller homes, 5.64 rooms against 6.36, with more debt, 0.68 of the value of the home against 0.55. A supply expansion instead gives young owners larger homes, 6.65 rooms, with the same debt. Credit buys earlier, smaller and more leveraged ownership; supply buys more space.

The PSID offers a partial check on this mechanism. Among childless first-time buyers aged 22 to 33, those who bought a home of four rooms or fewer were 14 percentage points less likely to have a first birth within six years (standard error 6.7). The gap goes in the model's direction, but without an instrument it is as consistent with selection, couples not planning children buying small, as with an effect of the home.

To isolate this margin from the response of prices, I raise ϕ to 1 in the 2007 stationary equilibrium, at the same price. Ownership at ages 25 to 34 rises from 58% to 85%. Figure 18 shows that the new owners are young and buy small homes: at ages 22 to 25, the share owning a home of 2 or 4 rooms rises from 12% to 49% (panel a). The share owning a home of 6 rooms or more rises by one percentage point among the youngest and falls slightly at older ages (panel b). Births move slightly earlier, and completed fertility is unchanged (panel c). The young use the credit to own the small homes they would otherwise have rented. Credit changes who owns and when, not the cost of a child's space.

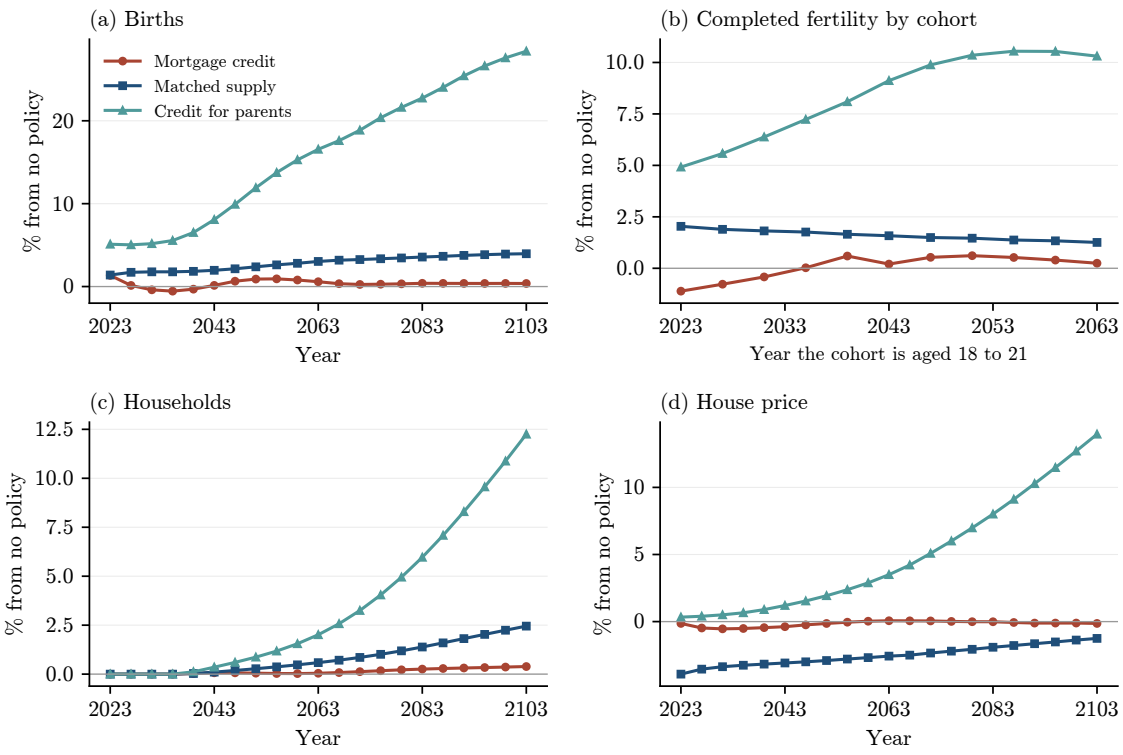


Figure 19: Mortgage credit, housing supply and credit for parents

Notes: % differences from the no-policy path from the same 2023 state. Births in each four-year period (panel a), completed fertility of the cohort aged 18–21 in the year shown (panel b), number of households (panel c) and house price (panel d). Housing supply: the supply curve shifted out by 4.35%. Credit for parents: LTV 100 for households with a child at home.

6.4 Credit for parents

If credit fails because it reaches households whose fertility was not held back, credit tied to children should do better. I raise the loan-to-value ratio to 1 only for households with a child at home. This lowers the cost of the child rather than the cost of waiting. Births in 2023 rise by 5.1%, several times as much as under credit for all households, at nearly the same price, 0.3% higher (Figure 19). The completed fertility of the 2023 cohort rises by 4.9%, and the share with a first birth by 4.1% (Table 5). Births keep rising, by 17% by 2063. As the larger cohorts demand more space, the price rises, by 3.5% by 2063 and 33% in the long run, when there are 36% more households.

The policy works through two channels, which the experiment does not separate. First, a birth unlocks ownership without a down payment, so the policy rewards the birth itself. Second, a parent who owns can borrow up to the full value of the home in every period with a child at home, a buffer in the state that carries the commitment.

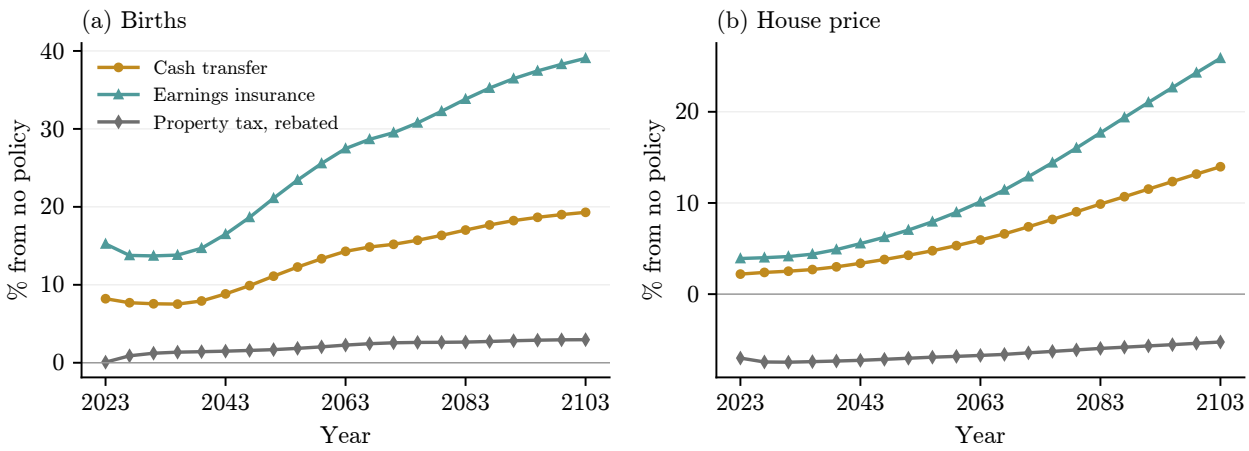


Figure 20: Cash, insurance and a property tax

Notes: % differences from the no-policy path from the same 2023 state, births in each four-year period (panel a) and house price (panel b).

6.5 Cash, insurance and a property tax

I compare the housing policies with three policies that work through households' resources rather than through housing. *Earnings insurance* pays every working household a quarter of the shortfall of its earnings below the median earnings state at its age. *Cash transfer* pays the expected value of that insurance at each age to every working household. Neither is financed within the model, so neither is a comparison at equal fiscal cost. *Property tax*: the annual tax rises from 1.06% to 2% of value, and the revenue is rebated equally to all households.

Transfers raise births far more than any of the housing policies (Figure 20). Cash raises births in 2023 by 8.2% and the completed fertility of the 2023 cohort by 7.8%; insurance by 15.2% and 14.0%. The difference, 6 to 7 percentage points, is the value of insuring the downside of earnings rather than paying the same amount to everyone: a child is a long commitment, and insurance makes it safer. Both raise the price, by 24% and 48% in the long run, as the larger population demands more space.

Insurance is a partial version of lower earnings risk, which gives the benchmark. Lowering the standard deviation of the four-year earnings shock from 0.48 to 0.35 from 2023 on, with mean earnings unchanged, raises births in 2023 by 11.3% and the completed fertility of the 2023 cohort by 16.0%, from 1.64 to 1.91. The additional demand for family space raises the price, by 11% by 2063 and 49% in the long run.

The property tax lowers the price by 7.0% in 2023, more than the supply expansion, but births do not move (+0.1%) and the completed fertility of the 2023 cohort rises by only 1.4%. The tax lowers the price because it raises the cost of holding housing, so the per-period cost of a room rises: rents are 14.5% higher in 2023. The rebate returns the revenue per household, not per room, so a family that needs more space pays more. A lower price is not cheaper space.

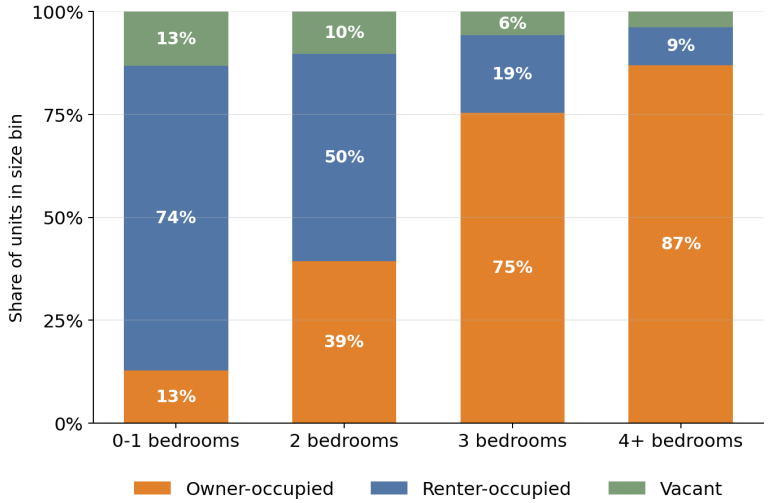
7 Conclusion

I build a life-cycle model of fertility and housing in which households choose when to have children, how much space to occupy, whether to rent or own, and how much to save. Children need space, the largest homes must be bought with a down payment, and earnings are uninsurable. House prices and rents are determined in equilibrium, and the children born today become households a generation later. I calibrate the model to the United States in 2007, take the fall in fertility since then as given, and study housing policies from the 2023 economy.

Mortgage credit for all households mainly moves births earlier. Young households use it to buy the small homes they would otherwise rent, and the completed fertility of the 2023 cohort falls by 1.1%.[‡] Policies that lower the cost of a child raise completed fertility: more housing supply, credit for parents, by 4.9%, and insurance against low earnings, by 14.0%, against 7.8% for cash. A rebated property tax lowers the house price but raises rents, and a lower price is not cheaper space. In the long run births per household return to replacement, so the gains show up as more households and build up over a generation. Policy raises births when it lowers the cost of a child rather than the cost of a home, and housing policy for births must be judged by its long-run effects.

Several features of the model matter for the size of these effects. The model overstates homeownership among young households, which places too many of them near the margin where credit acts. A common decline in the preference for children overstates childlessness in 2023 and misses the postponement of first births, and it makes the house price fall after 2007, while in the data it rose. Children ever born by age 25 are below the data. Credit for parents does not price default or mortgage-insurance costs, and the transfers are not financed. Landlords and developers are absentee, so rental income and capital gains on rental housing leave the economy.

Figure 21: Tenure of the Housing Stock by Number of Bedrooms (AHS 2023)



Note: This figure shows the tenure of housing units by number of bedrooms. Each bar divides the housing units with the stated number of bedrooms into owner-occupied units, units rented for cash, and vacant units. Units occupied without payment of cash rent and units whose occupants have a usual residence elsewhere are excluded. Weighted by the housing-unit weight. Source: American Housing Survey 2023, national public-use file.

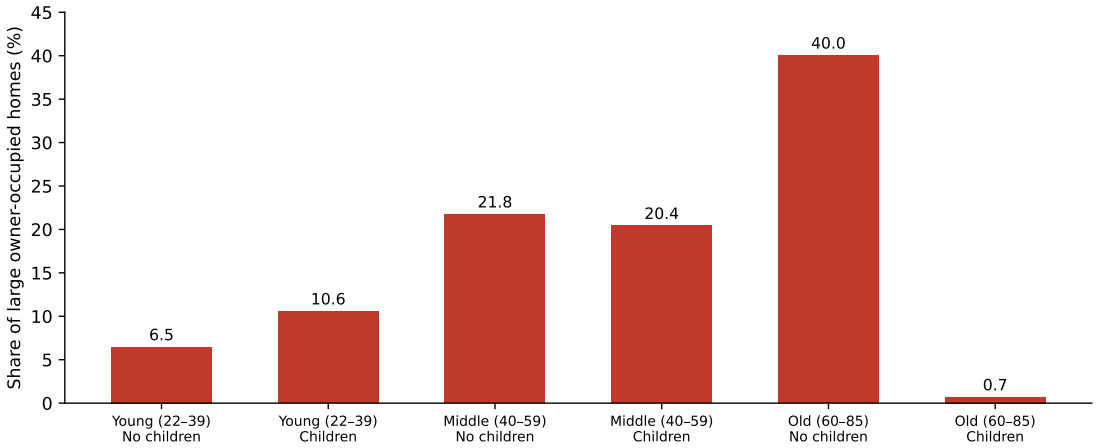
A Housing Stock and Family Space

This Appendix documents two facts about the housing stock that complement the event study of Section 2: large homes are mostly owned, and most large owned homes are held by older households.

A.1 Large homes are owned

Figure 21 divides the 2023 housing stock in the American Housing Survey (AHS) by number of bedrooms and tenure. Of the homes with four or more bedrooms, 87 percent are owner-occupied and 9 percent are rented. Of the homes with at most one bedroom, 13 percent are owner-occupied and 74 percent are rented. In rooms, the unit of the rest of the paper, 43 percent of owner-occupied homes have seven or more rooms, against 7.5 percent of rented homes. The gap holds among families: owners with children occupy 6.8 rooms on average and renters with children 5.2, and 48.7 percent of renter households with children live in homes with at most two bedrooms, against 7.4 percent of owner households with children. These shares describe the homes households occupy rather than the homes on offer, but they show that, in the existing stock, a family that wants a large home typically buys one.

Figure 22: Who Holds the Large Owner-Occupied Homes (ACS 2023)



Note: This figure shows who holds the large owner-occupied homes. Each bar is the share of owner-occupied homes with six or more rooms held by households in the group; the six shares sum to 100 percent. Groups are defined by the age of the householder and by whether an own child of the householder younger than 18 lives in the household. Householders aged 22 to 85 in 42 large metropolitan areas, weighted by the household weight; 289,109 households.

A.2 Older households hold the large homes

Figure 22 divides the owner-occupied homes with six or more rooms in 42 large metropolitan areas in the 2023 American Community Survey (ACS) by the age of the householder and by whether a child of the householder younger than 18 lives in the home. Households aged 60 to 85 without children at home hold 40.0 percent of these homes, and households aged 40 to 59 without children at home another 21.8 percent. Households aged 22 to 39 with children hold 10.6 percent. Across all ages, 68.3 percent of the large owner-occupied homes house no child of the householder, close to the 72.9 percent of all households in these areas without a child at home. The large homes are therefore not held disproportionately by households without children; they are held by the old. I call this concentration the intergenerational allocation of housing. For a young family, it means that the existing large homes become available mainly as older households sell or leave them. The figure says nothing about the price at which older owners would sell, or about whether young families would have more children if they held these homes.

A.3 Ownership of recent parents

The same pattern appears in the cross-section. In the 2005–2006 ACS, household heads aged 30 to 55 whose oldest child is under four are 12.8 percentage points more likely to own than heads without children, a moment the calibration also targets. Unlike the PSID estimates, this gap compares different households at a point in time.

A.4 Data

The stock facts use the national public-use household file of the 2023 American Housing Survey, weighted by the housing-unit weight. Tenure has three occupied categories: owned, rented for cash, and occupied without payment of cash rent. Shares within a tenure use all units of that tenure as the denominator; the share of seven-plus-room homes that are rented instead uses all occupied units of that size. It is 8.2 percent when the denominator contains owned and cash-rented units and 8.1 percent when it also contains units occupied without cash rent and units with unreported tenure. Bedrooms and total rooms are different measures. I report bedrooms for comparison with other work and rooms because rooms are the PSID and ACS outcome and the unit of the model. Mean rooms among households with children are computed over households with at least one child present. A cross-section of occupied homes does not identify the units offered for rent or the choices families would make at other prices.

B Data and Empirical Measurement

B.1 PSID first-birth event studies

Sample. The PSID data are the family and individual files for 1968–2023. An observation is a person-year in which the person is a current member of a family unit and at least 18 years old. The year of first birth is the earliest birth year of a biological child in the person's relationship-to-child history, which records up to twenty children. The comparison group consists of persons whose full history reports no children; persons with incomplete or unknown histories are dropped. A parent enters only if the first birth occurs no earlier than the first year in which the person is observed as an adult in the PSID. All estimates use the PSID longitudinal individual weight and so include only sample persons with a positive weight.

The household sample keeps women who are the reference person or spouse of a household containing a single current family unit, and keeps one woman per household and year, the reference person first. The sample of adults heading a household at baseline keeps treated adults, women and men, who were reference person or spouse in the window two to three years before the first birth, and retains all of their person-years. Eligible childless controls are retained without this treated-sample restriction. The complementary sample keeps treated adults who were in the PSID but not reference person or spouse in that window. Each sample compares first-time parents with eligible never-parents using the same weights and entry rule. The main text leads with the sample defined by household position before the birth, which is also the sample definition used for the calibration's room-response target. The other definitions show how the estimates depend on selecting a household in the years around parenthood.

For the room response used in calibration, I further restrict observations to the common support

of the empirical panel and the model measurement rules. This retains 112,148 person-years, 3,131 first-time parents and 5,954 childless controls. On this support, the room increase three to four years after birth is 1.44 (standard error 0.05). The unrestricted baseline-head sample gives 1.47 (0.05), as reported in the tables below. The difference is the observation support, not a different event-time reference window or birth definition. The tenure and moving regressions use their own available observations and do not impose this room-specific restriction.

Housing measures. All four housing items are taken from the official year-specific PSID family variables and merged to the interview year in which they were reported. Rooms are the total rooms in the dwelling except bathrooms, from the PSID question "How many rooms do you have (for your family) not counting bathrooms?". Codes that do not denote a count (9 through 1984; 99 in 1985–1993; 98 and 99 from 1994) are set to missing, and a reported zero is retained. Ownership equals one if the family owns the dwelling and zero if it rents; other arrangements are missing. The mobility item asks whether the family moved since a fixed date: the spring of the previous survey year in the annual waves, and January 1 of the previous survey year in recent waves. It therefore records a move within an interval of about one year in the annual waves and about two years in the biennial waves. The reason for a move is the first-mentioned reason. "More space" is the PSID category for expansion of housing (more space, more rent, a better place); "neighborhood" is the category for neighborhood-related reasons (a better neighborhood, going to school, being closer to friends or relatives). Unknown or unreported answers are missing, and code 9 of the reason item (unknown before 2019, homelessness in 2019) is treated as missing throughout. An observation contributes to an outcome only when that outcome is reported; unreported outcomes are never coded as zero.

Dating housing outcomes to the interview at which they are reported keeps the outcome and first-birth clocks aligned.

Estimator. Event time is the survey year minus the year of first birth, grouped into the windows ≤ -8, $-7/-6$, $-5/-4$, $-3/-2$, $-1/0$, $+1/+2$, $+3/+4$, $+5/+6$, $+7/+8$, $+9/+10$ and $\geq +11$. Two-year windows are needed because the PSID became biennial after 1997: from the 1999 birth cohort on, no cohort is observed at both single years -2 and $+3$, so a single-year reference period is observed for only part of the cohorts. The omitted window is $-3/-2$, and each birth cohort must be observed in it; cohorts without an observation in the reference window are excluded (first births in 1968–1970 and 1985–1986 in the household sample, none in the sample fixed at baseline). I estimate cohort-specific event paths relative to the comparison group and average them with cohort-share weights, following Sun and Abraham (2021). The regressions include person and survey-year fixed effects and complete sets of indicators for age and years of education. Standard errors are clustered by person and account for the estimation of the cohort shares.

Robustness. Table 6 collects alternative rooms estimates for the sample of the main text. With the year before the birth in the reference window ($-2/-1$), rooms are 1.12 higher two to three years after the birth. Moving the reference window earlier gives 1.47 and 1.35 at three to four years, because rooms change little before the birth in this sample; the increase measured from years $-3/-2$ is 1.47, 1.47 and 1.42 across the three reference windows. Using the harmonized room measure, aligned to the interview at which it was reported, instead of the official year-specific items changes the estimate by less than 0.01. Using the latest birth cohort instead of the childless as comparison group leaves it unchanged. Requiring the first birth to follow the first observation as reference person or spouse does not bind in this sample, whose treated adults are reference person or spouse two to three years before the birth.

Table 6: Rooms Around the First Birth: Alternative Specifications (PSID)

	(1) Baseline	(2) Reference $-2/-1$	(3) Reference $-5/-4$	(4) Reference $-7/-6$	(5) Harmonized room measure	(6) Latest cohort as controls
Change in rooms	1.47^{***}	1.12^{***}	1.47^{***}	1.35^{***}	1.46^{***}	1.46^{***}
	(0.05)	(0.04)	(0.06)	(0.07)	(0.05)	(0.05)
Reference window	$-3/-2$	$-2/-1$	$-5/-4$	$-7/-6$	$-3/-2$	$-3/-2$
Reported window	$+3/+4$	$+2/+3$	$+3/+4$	$+3/+4$	$+3/+4$	$+3/+4$
Reference-window mean rooms	4.69	4.87	4.81	5.06	4.70	4.69
Observations						
N	117,853	117,853	116,224	114,460	113,225	117,853
N individuals	9,310	9,310	9,263	9,205	9,235	9,310
N first-time parents	3,302	3,302	3,255	3,197	3,229	3,275

Standard errors in parentheses. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This table reports alternative estimates of the change in rooms around the first birth for the sample of the main text: adults, women and men, who were reference person or spouse two to three years before the birth. Column (1) is the baseline of Table 7, column (2). Each other column changes one element. Columns (2) to (4) move the reference window, holding the sample fixed. Column (5) uses the harmonized room measure, aligned to the interview at which it was reported, instead of the official year-specific items. Column (6) uses the latest first-birth cohort as the comparison group and keeps never-parents in the sample. "Reference window" is the omitted window and "Reported window" the reported one, both in years relative to the first birth.

B.2 ACS matched pseudo-panel

The ACS exercise uses the one-year samples for 2005–2019 in all states and the District of Columbia. The sample consists of women aged 25 to 45. Because the ACS contains no birth histories, the year of first birth is inferred from the age of the oldest child linked to the woman in the household roster, and a woman's housing before that year is represented by childless women surveyed in the corresponding earlier years who share her demographic matching cell. Housing outcomes do not enter the matching. Rooms are capped at nine, and codes that do not denote a count are missing; bedrooms are capped at five; ownership equals one for owned units and zero for rented ones. The regression includes event-time indicators from -5 to $+10$ with year -2 omitted, and state, age and survey-year fixed effects; it is weighted by the matching weights, with standard errors clustered by

source household because a household can supply several matched records. The estimation sample has 14.2 million matched records from 2.8 million source households.

Between one year before and three years after the birth, rooms rise by 0.49 (0.01), bedrooms by 0.34 (0.004), and the ownership rate by 3.3 percentage points (0.2). Dropping the age fixed effects, or all fixed effects, raises the room estimate to 0.70 and 0.71 and the ownership estimate to 10.6 and 10.7 points; dropping the state and survey-year effects gives 0.53 rooms and 4.3 points. The pre-birth observations are constructed from other women, so the design assumes that matched childless women represent the housing future mothers would have had before the birth. The roster does not record children living elsewhere or distinguish biological from other children.

B.3 Sibling-composition designs

The ACS sibling designs use women aged 21 to 35 in the 2005–2019 one-year samples whose oldest linked child, if any, is younger than 18. Children are linked to mothers through the household roster, which records children living with the mother, not all children borne, and does not separate biological from step or adoptive children. The regressions control for complete sets of indicators for the mother's age at the event, her race, the survey year and the event age, are weighted by the person weight, and cluster standard errors by household.

In the first design, the sample is mothers whose oldest linked child is aged 0 to 5 (857,607 mothers). The instrument indicates that the two oldest children are recorded at the same age, and the treatment is a second resident child. Ages are recorded in whole years without quarter of birth, so the instrument is an age-only proxy for a twin birth. The first stage is 0.66 (F statistic above 77,000). The reduced form is 0.33 rooms (0.02), and the implied effect of a second child is 0.50 rooms (95 percent interval 0.45 to 0.55), 0.33 bedrooms (0.31 to 0.36), and 4.7 percentage points in ownership (3.3 to 6.1).

In the second design, the sample is mothers whose two oldest linked children differ in age and whose second child is aged 0 to 5 (656,986 mothers); it is not restricted to mothers with exactly two children. The instrument indicates that the two oldest children have the same sex, and the treatment is a third resident child. The first stage is 0.033 ($F = 715$). Mothers of same-sex pairs live in 0.038 fewer rooms (0.005) and 0.034 fewer bedrooms (0.003), and their ownership rate does not differ. Scaled by the first stage, a third child would reduce the dwelling by 1.15 rooms (95 percent interval 0.83 to 1.47). Children of the same sex can share a bedroom, so sibling sex composition may change the space a family needs without changing its size. That possibility would violate the exclusion restriction for housing outcomes. The data cannot test it formally, and I do not interpret this estimate as the effect of a third child. The same two designs are too imprecise in the PSID to add information: same-age births are rare in its smaller sample, and the sex-composition first stage is weak.

B.4 Sample comparisons

Tables 7 and 8 compare two sample definitions before imposing the calibration's common-support restriction. The first keeps women who are reference persons or spouses in each year. The second keeps treated adults, women and men, who held that position two to three years before the birth and follows them thereafter; eligible childless controls are retained. The second is the household-position definition used in the main text. The room table's 1.47 estimate uses its full observation support, whereas the main-text room figure and the calibration use the restricted-support estimate of 1.44.

Figures 23 and 24 display the paths under both definitions. Fixing household position before the birth removes the pronounced pre-birth increase in rooms visible when the sample is reselected in each interview. It also yields larger post-birth room and ownership responses. The comparison makes household formation and sample selection relevant to the measurement of housing around parenthood; it does not isolate a causal effect of becoming a household head.

Both partners of a couple can appear in the sample. Clustering standard errors by household instead of by person leaves the estimates unchanged and raises the standard errors by at most 2.4%.

B.5 House value around the first birth

The rise in rooms could reflect a move to a cheaper location, where space costs less, rather than a larger expenditure on housing. To check this, I estimate equation (2.1) for the log of house value, the log of rooms and the log of house value per room among owners, and for a move across state lines among all households. House values are reported by the household and expressed in 2022 dollars. Table 9 reports the estimates three to four years after the birth. Among households that already owned in the reference window, house value rises by 30 percent and rooms by 17 percent, so value per room rises by 11 percent. The coefficients on value per room before the reference window are small and not significant. Moves across state lines become less frequent after the birth. Households therefore do not obtain space by moving to places where it is cheaper. Value per room reflects both location and the quality of the home, so the estimates do not rule out a move to a suburb with similar prices per room.

Table 7: Rooms Around the First Birth (PSID)

	(1) Women who are reference person or spouse in each year	(2) Adults who were reference person or spouse two to three years before the birth
Years relative to first birth		
−8 or earlier	−0.50***	0.15+
	(0.12)	(0.08)
−7 and −6	−0.50***	0.04
	(0.10)	(0.06)
−5 and −4	−0.34***	−0.03
	(0.06)	(0.04)
−3 and −2	reference window	
−1 and 0	0.41***	0.61***
	(0.05)	(0.04)
+1 and +2	0.82***	1.19***
	(0.06)	(0.04)
+3 and +4	1.03***	1.47***
	(0.07)	(0.05)
+5 and +6	1.17***	1.62***
	(0.08)	(0.06)
+7 and +8	1.20***	1.64***
	(0.09)	(0.07)
+9 and +10	1.25***	1.83***
	(0.10)	(0.08)
+11 or later	1.16***	1.85***
	(0.13)	(0.09)
Reference-window mean rooms	4.70	4.69
Observations		
N	63,338	117,853
N individuals	5,309	9,310
N first-time parents	3,655	3,302

Standard errors in parentheses. + $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This table reports interaction-weighted event-study estimates (Sun and Abraham, 2021) of the change in rooms relative to years −3 and −2, in two-year windows of event time, with standard errors clustered by person. The comparison group consists of adults with no children in their full relationship history. All regressions include person and survey-year fixed effects and indicators for age and years of education, and use PSID longitudinal weights. Rooms are the total rooms in the dwelling reported at the interview. Column (1) excludes first births in 1968–1970 and 1985–1986, which have no observation in the reference window. N individuals counts first-time parents and childless controls.

Table 8: Ownership and Moving Around the First Birth (PSID)

	(1) Owns the dwelling	(2) Moved since last interview	(3) Moved for more space	(4) Moved for neighborhood	(5) Share of moves for more space	(6) Share of moves for neighborhood
A. Women who are reference person or spouse in each year						
−1 and 0	0.11***	−0.04**	0.014	−0.004	0.054**	−0.017
	(0.01)	(0.02)	(0.009)	(0.006)	(0.017)	(0.012)
+1 and +2	0.16***	−0.12***	0.034***	−0.005	0.140***	−0.011
	(0.02)	(0.02)	(0.009)	(0.006)	(0.021)	(0.014)
+3 and +4	0.16***	−0.16***	0.020*	−0.010+	0.133***	−0.017
	(0.02)	(0.02)	(0.010)	(0.005)	(0.023)	(0.014)
+5 and +6	0.14***	−0.16***	0.007	0.002	0.108***	0.022
	(0.02)	(0.02)	(0.010)	(0.006)	(0.026)	(0.018)
Reference-window mean	0.41	0.58	0.070	0.032	0.122	0.054
N	61,229	63,817	62,813	62,813	19,456	19,456
N individuals	5,238	5,315	5,282	5,282	4,327	4,327
N first-time parents	3,619	3,656	3,640	3,640	3,105	3,105
B. Adults who were reference person or spouse two to three years before the birth						
−1 and 0	0.16***	−0.07***	0.015*	−0.009*	0.050***	−0.022**
	(0.01)	(0.01)	(0.006)	(0.004)	(0.013)	(0.009)
+1 and +2	0.25***	−0.17***	0.021***	−0.008*	0.114***	0.005
	(0.01)	(0.01)	(0.006)	(0.004)	(0.017)	(0.011)
+3 and +4	0.27***	−0.21***	0.010	−0.009**	0.142***	−0.002
	(0.01)	(0.01)	(0.006)	(0.004)	(0.020)	(0.013)
+5 and +6	0.26***	−0.21***	0.009	−0.003	0.147***	0.012
	(0.01)	(0.01)	(0.006)	(0.004)	(0.022)	(0.015)
Reference-window mean	0.43	0.54	0.064	0.030	0.123	0.058
N	111,272	118,875	116,697	116,697	27,462	27,462
N individuals	9,070	9,346	9,245	9,245	5,488	5,488
N first-time parents	3,142	3,323	3,255	3,255	2,184	2,184

Standard errors in parentheses. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This table reports the change in ownership and moving in windows of years relative to the first birth, relative to years −3 and −2, with the same estimator, comparison group, controls, weights and reference window as Table 7. The reference-window mean is the weighted mean among first-time parents in years −3 and −2. Each outcome uses its own sample of non-missing observations. "Moved for more space" equals one if the household moved since the previous interview and gave expansion of housing (more space, more rent, a better place) as the first reason, and zero otherwise; "moved for neighborhood" is defined in the same way for neighborhood-related reasons (a better neighborhood, going to school, being closer to friends or relatives). Columns (5) and (6) restrict the sample to household-years with a move and therefore condition on an outcome of the birth.

Table 9: House Value Around the First Birth (PSID)

	All owners			Owners in −3/−2			All households
	(1) Log house value	(2) Log rooms	(3) Log value per room	(4) Log house value	(5) Log rooms	(6) Log value per room	(7) Moved across state lines
Years +3 and +4	0.230***	0.146***	0.084***	0.265***	0.157***	0.108***	−0.021**
	(0.024)	(0.011)	(0.021)	(0.025)	(0.012)	(0.023)	(0.007)
Reference-window mean	228.3	5.96	38.9	228.4	5.94	39.3	0.089
Observations							
N	65,498	65,498	65,498	44,320	44,320	44,320	112,027
N individuals	6,501	6,501	6,501	4,712	4,712	4,712	8,961
N first-time parents	2,831	2,831	2,831	1,042	1,042	1,042	3,282

Standard errors in parentheses. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This table reports estimates of β_k in equation (2.2) for the window three to four years after the first birth, relative to two to three years before it. Sample, estimator, weights and clustering as in Figure 1, without the common-support restriction. Columns (1) to (6) keep owners with a positive house value and rooms; columns (4) to (6) keep first-time parents who owned at every interview two to three years before the birth. Each outcome uses its available observations. The reference-window mean is the weighted level among first-time parents in years −3 and −2: house value and value per room in thousands of 2022 dollars, rooms, and the share moving across state lines. A move across state lines compares the state of residence with the previous interview, one or two years earlier.

C Household Optimality Conditions

In this Appendix, I derive the conditions that Section 3.4 uses: the choices within a housing option, the Euler equation for the bond and the borrowing multiplier ζ_t, the effect of credit on the value of a household and on fertility, and the two results in the footnotes of that Section. Throughout, n is the number of children at home after the birth outcome of period t, and the budget constraint (3.7) of the chosen option defines c_t as a function of b_{t+1} and, for a renter, of h_t^R.

Choices within an option. Within option j of (3.6), renting or owning $h_t^O \in \mathcal{H}$, the household chooses b_{t+1} and, if it rents, h_t^R, subject to the borrowing limit (3.2), with $h_t^O = 0$ for a renter, and to the cap $h_t^R \leq \bar{h}^R$. Let $\zeta_t \geq 0$ and $\nu_t \geq 0$ be the multipliers on the two constraints. By (3.7), $\partial c_t / \partial b_{t+1} = -1$ in every option and $\partial c_t / \partial h_t^R = -r_t$ for a renter. In (3.1), the marginal utilities of consumption and of rented space are the same power of the composite in brackets, times α_0/c_t and $(1-\alpha_0)/(h_t^R - \bar{h}(n))$, so the condition $\partial u / \partial h_t^R = r_t \, \partial u / \partial c_t + \nu_t$ gives, below the cap,

$$r_t\big(h_t^R - \bar{h}(n)\big) = \frac{1-\alpha_0}{\alpha_0}\, c_t. \tag{C.1}$$

A renter spends the share α_0 of $c_t + r_t(h_t^R - \bar{h}(n))$ on consumption and $1-\alpha_0$ on space above the requirement. With $b_t = b_{t+1} = 0$ and no home to sell, this spending is $y_t + T_t - r_t \bar{h}(n)$, and substituting the two shares into (3.1) gives (3.10). At the cap, $h_t^R = \bar{h}^R$ and $\nu_t = \partial u / \partial h_t^R - r_t \, \partial u / \partial c_t \geq 0$. An owner's space is fixed within the option, so its only choice is the bond.

The bond. The first-order condition for b_{t+1} is the Euler equation

$$\frac{\partial u}{\partial c_t} = \beta\,\mathbb{E}_t\Big[s_a \frac{\partial V_{t+1}}{\partial b_{t+1}} + (1-s_a)\,B'(x_{t+1})\Big] + \zeta_t, \qquad \zeta_t\big(b_{t+1} + \phi P_t h_t^O\big) = 0, \tag{C.2}$$

where $h_t^O = 0$ for a renter and $B'(x_{t+1}) = \theta_0(\theta_1 + x_{t+1})^{-\sigma}$. The multiplier ζ_t is the gain, in period-t utility, from borrowing one more unit in period t and repaying R_b in period $t+1$: the marginal utility of consumption today net of the discounted expected marginal value of the bond tomorrow. It is positive only at the limit. The estate $x_{t+1} = b_{t+1} + (1-\mu)P_t h_t^O$ contains the bond position itself, so $\partial x_{t+1}/\partial b_{t+1} = 1$ and the bequest term carries no R_b; the survival term does, through the budget of period $t+1$.

For Gumbel shocks with scale κ, the expected maximum over options is the logsum in (3.5) and (3.8), whose derivative with respect to the value of one option is the probability of that option:

$$\frac{\partial}{\partial V^k}\,\kappa \log \sum_j \exp\big(V^j/\kappa\big) = \frac{\exp\big(V^k/\kappa\big)}{\sum_j \exp\big(V^j/\kappa\big)}. \tag{C.3}$$

The household observes the shocks before it chooses, so a marginal change in a state moves the value of the chosen option only; at the shocks where the choice switches, the two options have the same value. The derivative of the expected maximum with respect to a state is therefore the probability-weighted average of the derivatives of the option values. When $\sigma \geq 1$, the value of an option whose feasible set shrinks to a point tends to $-\infty$, since $u \to -\infty$ as $c_t \to 0$, so its probability tends to zero and the logsum is continuous where an option enters or leaves the feasible set. Within an option j of period $t+1$, b_{t+1} enters only the budget, with $\partial c_{t+1}/\partial b_{t+1} = R_b$, so $\partial V_{t+1}^j/\partial b_{t+1} = R_b\,\partial u/\partial c_{t+1}$ at the option's consumption, whether or not a constraint binds in that period. Writing $\frac{\partial u}{\partial c_{t+1}}(j, n')$ for this marginal utility with n' children at home, and applying (C.3) to (3.8) and to (3.5),

$$\begin{aligned}\frac{\partial V_{t+1}}{\partial b_{t+1}} = R_b\Big[\big(1 - \pi_{a+1}\pi^F_{t+1,a+1}\big)\sum_j \pi^H_{t+1}(j \mid n_{t+1})\,\frac{\partial u}{\partial c_{t+1}}(j, n_{t+1}) \\ + \pi_{a+1}\pi^F_{t+1,a+1}\sum_j \pi^H_{t+1}(j \mid n_{t+1}+1)\,\frac{\partial u}{\partial c_{t+1}}(j, n_{t+1}+1)\Big],\end{aligned} \tag{C.4}$$

where n_{t+1} is the number of children at home at the start of period $t+1$ and $\pi_{a+1}\pi^F_{t+1,a+1}$ is the probability of a birth in that period, zero outside the fertile ages. The derivative is R_b times the marginal utility of consumption in period $t+1$, averaged over the birth outcome and the housing choice of that period; the cost ξ drops out. The expectation $\mathbb{E}_t$ in (C.2) then averages over z_{t+1} and the departure of children.

A birth and the multiplier. At the limit, ζ_t is the difference between the two sides of (C.2) at $b_{t+1} = -\phi P_t h_t^O$, and a birth changes both. In period t, it adds $r_t h_P$ to the spending a renter cannot

avoid, lowers an owner's space above the requirement by χh_P, and raises $e(n)$; for $\sigma > 1$, each of these raises $\partial u/\partial c_t$ at the given bond position. In period $t+1$ the same holds with probability $1-\lambda$, when the child is still at home, and the bequest term does not depend on n. Within an option,

$$\zeta_t(h_t^O, n+1) - \zeta_t(h_t^O, n) = \Big[\frac{\partial u}{\partial c_t}\Big]_n^{n+1} - \beta s_a \, \mathbb{E}_t\Big[\frac{\partial V_{t+1}}{\partial b_{t+1}}\Big]_n^{n+1}, \tag{C.5}$$

where $[\,\cdot\,]_n^{n+1}$ is the difference between the values with $n+1$ and with n children at home, at $b_{t+1} = -\phi P_t h_t^O$ in both. The first term is positive, and larger when spending in period t is small, since the marginal utility of consumption is convex in spending. The second is scaled by βs_a and by $1-\lambda$, averaged with the choice probabilities of period $t+1$, and evaluated at that period's spending, which is larger when earnings rise with age. The multiplier therefore rises for a young household with low spending today and rising earnings, and can fall for one that expects to spend less next period; the sign is ambiguous in general.

Credit. Consider a change in ϕ in period t alone. It enters the problem of an owner only through the borrowing limit, whose derivative with respect to ϕ is $P_t h_t^O$, so by the envelope theorem $\partial V_t^j/\partial\phi = \zeta_t(h^O, n) P_t h^O$ for owning $h^O \in \mathcal{H}$ and $\partial V_t^j/\partial\phi = 0$ for renting, where $\zeta_t(h^O, n)$ is the multiplier in the option of owning h^O with n children at home. Applying (C.3) to (3.8) gives the second line of (3.9). For the probability of trying, (3.5) and $V_{t,A}^F - V_{t,0}^F = \pi_a\big[V_t^H(n_t+1) - V_t^H(n_t) - \xi\,\mathbf{1}\{\text{first child}\}\big]$ give

$$\frac{\partial \pi_{t,a}^F}{\partial \phi} = \frac{\pi_{t,a}^F\big(1-\pi_{t,a}^F\big)}{\kappa}\,\frac{\partial\big(V_{t,A}^F - V_{t,0}^F\big)}{\partial\phi} = \frac{\pi_{t,a}^F\big(1-\pi_{t,a}^F\big)}{\kappa}\,\pi_a\left[\frac{\partial V_t^H(n_t+1)}{\partial\phi} - \frac{\partial V_t^H(n_t)}{\partial\phi}\right], \tag{C.6}$$

where $\kappa = \kappa_1$ before the first child and κ_C after; the cost ξ does not depend on ϕ and drops out. This is the first line of (3.9). If the change in ϕ is permanent, it also enters V_{t+1}, and the envelope theorem adds $\beta s_a \mathbb{E}_t\, \partial V_{t+1}/\partial\phi$ to the derivative of each option, which by (C.3) is the second line of (3.9) in period $t+1$, averaged over the choices of that period, and so on: the derivative becomes the sum of these terms over all future periods, discounted by β and weighted by survival and by the probabilities of the choices that lead to each option.

The bracket in (3.9) splits into two parts,

$$\begin{aligned}\frac{\partial V_t^H(n+1)}{\partial\phi} - \frac{\partial V_t^H(n)}{\partial\phi} = &\sum_{h^O\in\mathcal{H}} \big[\pi_t^H(h^O \mid n+1) - \pi_t^H(h^O \mid n)\big]\,\zeta_t(h^O, n)\,P_t h^O \\ &+ \sum_{h^O\in\mathcal{H}} \pi_t^H(h^O \mid n+1)\big[\zeta_t(h^O, n+1) - \zeta_t(h^O, n)\big]\,P_t h^O,\end{aligned} \tag{C.7}$$

where the first sum holds the multipliers at their values without the child and the second holds the probabilities at their values with the child. The first sum is positive when the child moves probability from renting, and from homes where the limit is slack, toward homes where $\zeta_t(h^O, n) P_t h^O$ is large. The second is positive when the child raises the multiplier in the homes the household is likely to

own with it, with the sign of (C.5). Both are small for a household that rents with high probability with and without the child.

The owner's cost of space. By (3.7), an owner that holds a unit of housing from period t to period $t+1$ pays P_t and $\tau^p P_t$ in period t. In period $t+1$ it pays maintenance $\delta_H P_{t+1}$ if it stays, or receives $(1-\delta_H-\mu)P_{t+1}$ if it sells, so apart from the selling cost the unit is worth $(1-\delta_H)P_{t+1}$ in period $t+1$, which the bond values at $(1-\delta_H)P_{t+1}/R_b$ in period t. A renter pays r_t in period t. By (3.4),

$$r_t - \Big[(1+\tau^p)P_t - \frac{(1-\delta_H)P_{t+1}}{R_b}\Big] = 0, \tag{C.8}$$

where the term in brackets is the owner's cost in period-t units. Apart from the selling cost, owning a unit of space costs the same as renting it, along a transition as well as in a stationary equilibrium.

The cost of a child's space and earnings. Let $\tilde{y}_t = y_t + T_t$, $k = r_t h_P$ and $\Theta_t = \alpha_0^{\alpha_0}(1-\alpha_0)^{1-\alpha_0} r_t^{\alpha_0-1}$. The first term of (3.10) is $(\Theta_t \tilde{y}_t)^{1-\sigma}/(1-\sigma)$ without children and $\big(\Theta_t(\tilde{y}_t - k)/e(1)\big)^{1-\sigma}/(1-\sigma)$ with one child, so the utility cost of a first child and its second derivative are

$$\begin{aligned} \Delta_t(\tilde{y}_t) &= \frac{\Theta_t^{1-\sigma}}{1-\sigma}\Big[\tilde{y}_t^{1-\sigma} - e(1)^{\sigma-1}(\tilde{y}_t - k)^{1-\sigma}\Big], \\ \Delta_t''(\tilde{y}_t) &= \sigma\,\Theta_t^{1-\sigma}\Big[e(1)^{\sigma-1}(\tilde{y}_t - k)^{-\sigma-1} - \tilde{y}_t^{-\sigma-1}\Big], \end{aligned} \tag{C.9}$$

where $\tilde{y}_t > k$. For $\sigma \geq 1$, $e(1)^{\sigma-1} \geq 1$ and $(\tilde{y}_t - k)^{-\sigma} > \tilde{y}_t^{-\sigma}$, so $\Delta_t'(\tilde{y}_t) = \Theta_t^{1-\sigma}\big[\tilde{y}_t^{-\sigma} - e(1)^{\sigma-1}(\tilde{y}_t - k)^{-\sigma}\big] < 0$ and $\Delta_t'' > 0$: the cost is decreasing and convex in earnings. The second derivative rises with k, since $\partial\Delta_t''/\partial k = \sigma(\sigma+1)\Theta_t^{1-\sigma} e(1)^{\sigma-1}(\tilde{y}_t - k)^{-\sigma-2} > 0$, and, for $\sigma > 1$, with r_t, since $\Theta_t^{1-\sigma}$ is proportional to $r_t^{(1-\alpha_0)(\sigma-1)}$. For a renter at the cap with and without the child, $c_t = \tilde{y}_t - r_t \bar{h}^R$ in both cases and the child lowers space above the requirement from $\bar{h}^R$ to $\bar{h}^R - h_P$, so

$$\Delta_t(\tilde{y}_t) = \frac{(\tilde{y}_t - r_t\bar{h}^R)^{\alpha_0(1-\sigma)}}{1-\sigma}\Big[(\bar{h}^R)^{(1-\alpha_0)(1-\sigma)} - e(1)^{\sigma-1}\big(\bar{h}^R - h_P\big)^{(1-\alpha_0)(1-\sigma)}\Big],$$

which for $\sigma > 1$ is positive, decreasing and convex in $\tilde{y}_t$ and rises with r_t and h_P.

The figures of Section 3.4. I compute Figures 6 and 7 from the model at the parameters of Table 1. The estimated values are $\beta_{\text{ann}} = 0.9357$, $\psi_0 = 0.1774$, $\gamma = 0.6457$, $\xi = 0.2946$, $\kappa_1 = 0.0505$, $\kappa_C = 0.1285$, $h_P = 0.8899$, $\chi = 1.1121$, $\kappa_T = 0.0131$ and $\theta_0 = 0.2435$, and the assigned parameters are those of the same table. In the 2007 stationary equilibrium at these values, the price is $P_0 = 0.7707$, the rebate is $T = 0.1879$ per household and there are $N_0 = 1$ households. Supply (3.3) is $H = 6.776\,P^{0.63}$. Each solve below is a stationary solve of the household problem and of the distribution of households at a given price, with the rebate set so that the government budget balances at that price (relative residual below 10^{-6}). None of these solves is a transition. The stationary rent is $r = \big[1 + \tau^p - (1-\delta_H)/R_b\big]P = 0.1698\,P$. The axes of the figures carry no numerical scale, and I report the values of the marked points below.

Figure 6. This figure is computed from (3.10) and involves no solve. With $\sigma = 2$, the cost of a first child is $\Delta(y) = \big[e(1)/(y + T - rh_P) - 1/(y + T)\big]/\Theta$, where $\Theta = \alpha_0^{\alpha_0}(1-\alpha_0)^{1-\alpha_0} r^{\alpha_0 - 1}$. The values are $\alpha_0 = 0.733$, $e(1) = 1.35^{0.7} = 1.234$ and $h_P = 0.890$, and both curves use the 2007 rebate, $T = 0.188$. The solid curve uses the 2007 rent, $r = 0.1308$, so that $rh_P = 0.116$. The dashed curve uses the rent at the long-run price $P_1 = 0.394$ of Figure 7, $r = 0.0669$, so that $rh_P = 0.060$. Earnings $y_L = 0.274$ and $y_H = 0.752$ are after-tax earnings per period at ages 22–25 in the first and third of the nine efficiency states, $z = 0.103$ and $z = 0.284$. In the 2007 solve, at least 98% of the renters of that age in these two states end the period at the borrowing limit. Over the plotted range, neither a childless renter nor a parent reaches the rental cap $\bar{h}^R = 6$ at either rent. The additional cost due to risk, $\frac{1}{2}[\Delta(y_L) + \Delta(y_H)] - \Delta(\bar{y})$, is 0.245 at the 2007 rent and 0.106 at the long-run rent; the braces mark these two differences.

Figure 7. Both panels come from 22 solves at given prices. Ten are at $\psi_0 = 0.1774$, at prices from 0.30 to 0.92. The other twelve are at $\psi_1 = 0.1472$ over the same range, three of them steps of the search for P_1. Here ψ_1 is the long-run level ψ' of the decay path (4.2), with $\rho = 0.35$. [ψ_1 is from a nearby point; refit at the final calibration.] At each price, $n(P;\psi)$ is births per entering household, with the bin of three or more children counted as 3.60 and replacement at $\bar{n} = 2.1$, and $h(P;\psi)$ is housing demand per household. The demand and fertility curves interpolate the solved points with monotone cubic splines, and supply is $H = 6.776\,P^{0.63}$. At A, $P_0 = 0.771$, $H = 5.750$, $N_0 = 1$ and $n = \bar{n}$. At A', preferences are ψ_1 at P_0 with N_0 households, and $n = 1.632$. At B, the long-run stationary equilibrium, $P_1 = 0.394$, $n = \bar{n}$ (births within 10^{-6} of replacement), $H = 3.769$, $N_1 = H/h(P_1;\psi_1) = 0.472$ and $T = 0.133$. From A to B, the price falls by 49%, the stock by 34% and the number of households by 53%. In panel (a), demand at A' is $N_0 h(P_0;\psi_1) = 5.698$, which is 0.9% below demand at A, so I mark A' at A. The arrows only indicate the direction of the adjustment.

D Robustness of the Policy Results

This appendix repeats the experiments of Section 6 with a slowly adjusting stock, at two supply elasticities.

D.1 A slowly adjusting stock

Here the housing stock adjusts slowly, as in Kaplan et al. (2020). Builders combine goods X_t with a fixed flow of permits $\bar{L}$ each period, $I_t = X_t^{\alpha} \bar{L}^{1-\alpha}$, so the marginal cost of a unit rises with this period's construction, not with the stock. The permit rent, $(1-\alpha)P_t I_t$, goes to the government, which spends it on goods households do not value. Setting $\alpha/(1-\alpha) = \eta$ and the

scale of construction to $\delta_H H_0$, construction and the stock follow

$$I_t = \delta_H H_0 P_t^{\eta}, \qquad H_t = (1 - \delta_H) H_{t-1} + I_t. \tag{D.1}$$

In a stationary equilibrium construction replaces depreciation, so the stock lies on the same curve $H = H_0 P^{\eta}$ as under free entry, and the 2007 equilibrium is the same. Along a transition only construction responds to the price within a period, so the stock responds with an elasticity of about $\delta_H \eta$, and the price absorbs most of a change in demand. The decline in the preference for children from 2007 to 2023 is fitted under this rule, so the 2023 state differs from the one in Section 6. Everything else is as in Section 6.1.

Table 10 reports the results. With $\eta = 0.63$, mortgage credit raises births by 1.7% in 2023, but the gain falls to 0.6% by 2063, and the completed fertility of the 2023 cohort falls by 0.2%. Ownership at ages 18 to 29 rises by 19 percentage points on impact. A 10% supply expansion lowers the price by 3.0% in 2023 and raises births by only 0.4%. Construction keeps adding space, rents are 4% lower by 2063, births are 3.2% higher, and the completed fertility of the 2023 cohort rises by 1.4%. Credit for parents raises births by 6.2% in 2023 and by 17% by 2063, and the completed fertility of the 2023 cohort by 5.9%. As the additional families demand more space, the house price rises, by 9% by 2063. Figure 26 shows the paths.

With $\eta = 1.5$ (panel B), the effects on births in 2023 and 2063 are close to those with $\eta = 0.63$. To raise births in 2023 as much as mortgage credit, the supply expansion must be 62%. It then raises births by 12% by 2063 and the completed fertility of the 2023 cohort by 5.6%, against −0.8% under credit.

[Problem: With $\eta = 1.5$ the paths are still well short of the terminal steady state in 2403.]

Table 10: Housing policies from 2023 with a slowly adjusting stock

	Births			Completed fertility	House price	
	2023	2063	2103	2023 cohort	2023	2063
A. Supply elasticity $\eta = 0.63$						
Mortgage credit	+1.7	+0.6	+0.9	−0.2	+0.2	+0.2
Housing supply, +10%	+0.4	+3.2	+5.1	+1.4	−3.0	−4.7
Credit for parents	+6.2	+17.1	+23.1	+5.9	+1.8	+9.4
B. Supply elasticity $\eta = 1.5$						
Mortgage credit	+1.9	+0.6	+0.0	−0.8	−0.1	+0.2
Housing supply, +10%	+0.3	+2.4	+3.8	+1.1	−2.1	−3.1
Matched supply, +62%	+1.9	+12.4	+20.6	+5.6	−11.1	−15.2
Credit for parents	+6.4	+17.2	+25.6	+7.5	+1.1	+6.0

Notes: % differences from the no-policy path from the same 2023 state, at the same supply elasticity, with the stock following (D.1). Births: births in the four-year period starting in the year shown. Completed fertility: children ever born to the cohort aged 18–21 in 2023. Without policy and with $\eta = 0.63$, period fertility is 1.64 in 2023, 1.80 in 2063 and 2.00 in 2103. Matched supply: the supply shift that raises births in 2023 as much as mortgage credit.

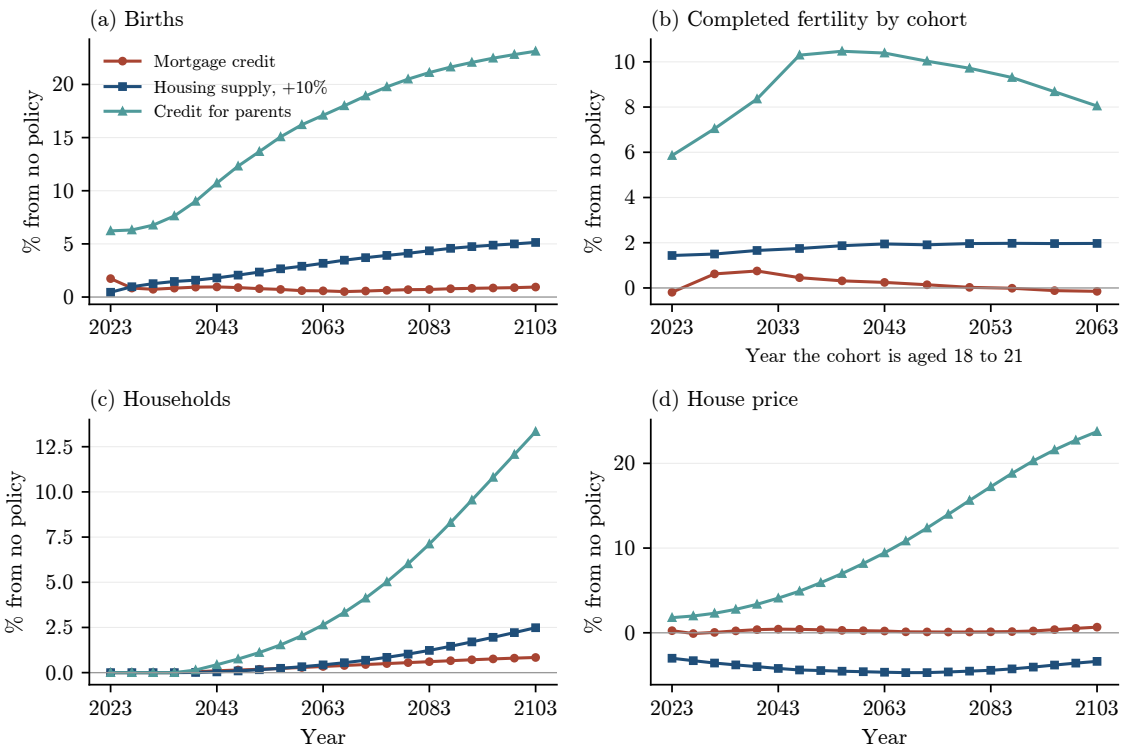


Figure 26: Mortgage credit, housing supply and credit for parents with a slowly adjusting stock

Notes: % differences from the no-policy path from the same 2023 state, $\eta = 0.63$, stock following (D.1). Births in each four-year period (panel a), completed fertility of the cohort aged 18–21 in the year shown (panel b), number of households (panel c) and house price (panel d). Housing supply: the supply curve shifted out by 10%.

E Solution Algorithms

Stationary equilibrium. I solve for the house price P and the rebate T, given parameters, a fixed preference for children and the payroll tax. The pension is set before the search: in a stationary population the age distribution follows from survival and entry, so the pension that balances the payroll tax is known in advance, and the balance is verified at the end.

1. Guess P and derive the rent r from the user cost.
2. Solve the household problem backward over age and compute the choice probabilities.
3. Construct the stationary distribution of households, iterating on the rebate until T times the number of households equals property-tax revenue. In the calibration the number of households is one and H_0 is set so that supply equals demand at P; in the counterfactuals H_0 is fixed and the number of households equals housing supply divided by mean housing per household.
4. Compute the renewal residual: adjusted births divided by 2.1, relative to the number of entrants, minus one. It is zero when the population reproduces itself.
5. Update P by bracketing the root of the renewal residual and repeat until the renewal and

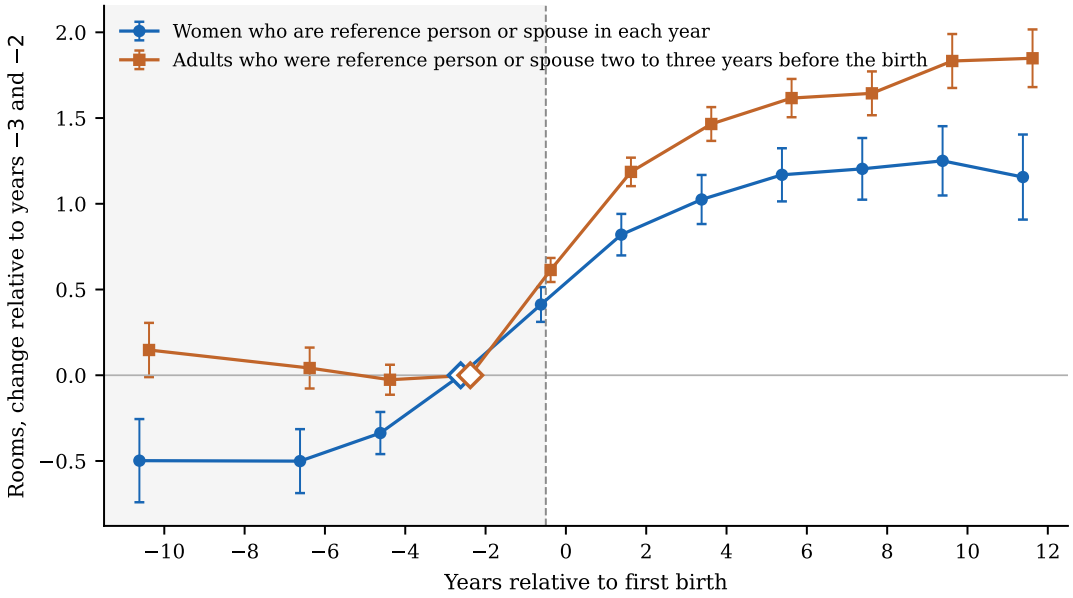
rebate residuals are below 10^{-6}. Re-solve at the selected price and require the same residuals, and verify that the pension budget balances to the same tolerance.

Transition. I solve for the paths $(P_t, m_t, T_t)_{t \geq t_0}$, given the inherited households and a preference believed permanent.

1. Solve the new stationary equilibrium; it supplies the terminal value function.
2. Guess the three paths and derive rents from (3.4).
3. Solve the household problem backward in calendar time from the terminal value.
4. Advance households and the queue of entrants forward from the inherited state, and aggregate housing demand and taxes at each date.
5. Compute the three residuals at every date. The loss is the largest residual across dates.
6. Update the paths jointly and repeat until the loss is below tolerance. Extend the horizon to check that early responses are stable.

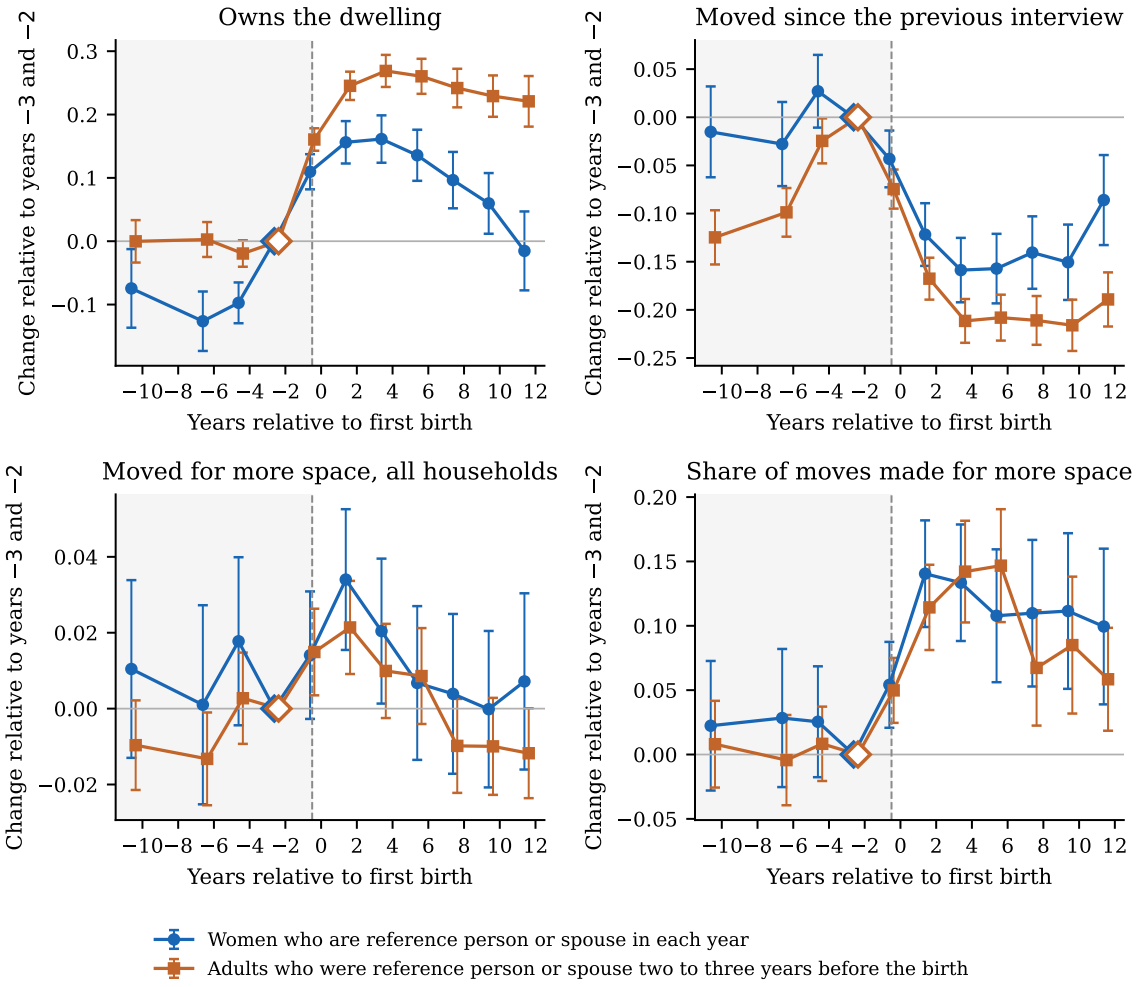
The transitions in the paper use 100 four-year periods, from 2007 (or 2023 for the policy experiments) to 2403, with tolerances of 2×10^{-4} on the housing market and 2×10^{-5} on the pension and the rebate, and 10^{-6} on the renewal condition of the terminal steady state.

Figure 23: Rooms Around the First Birth, Two Samples (PSID)



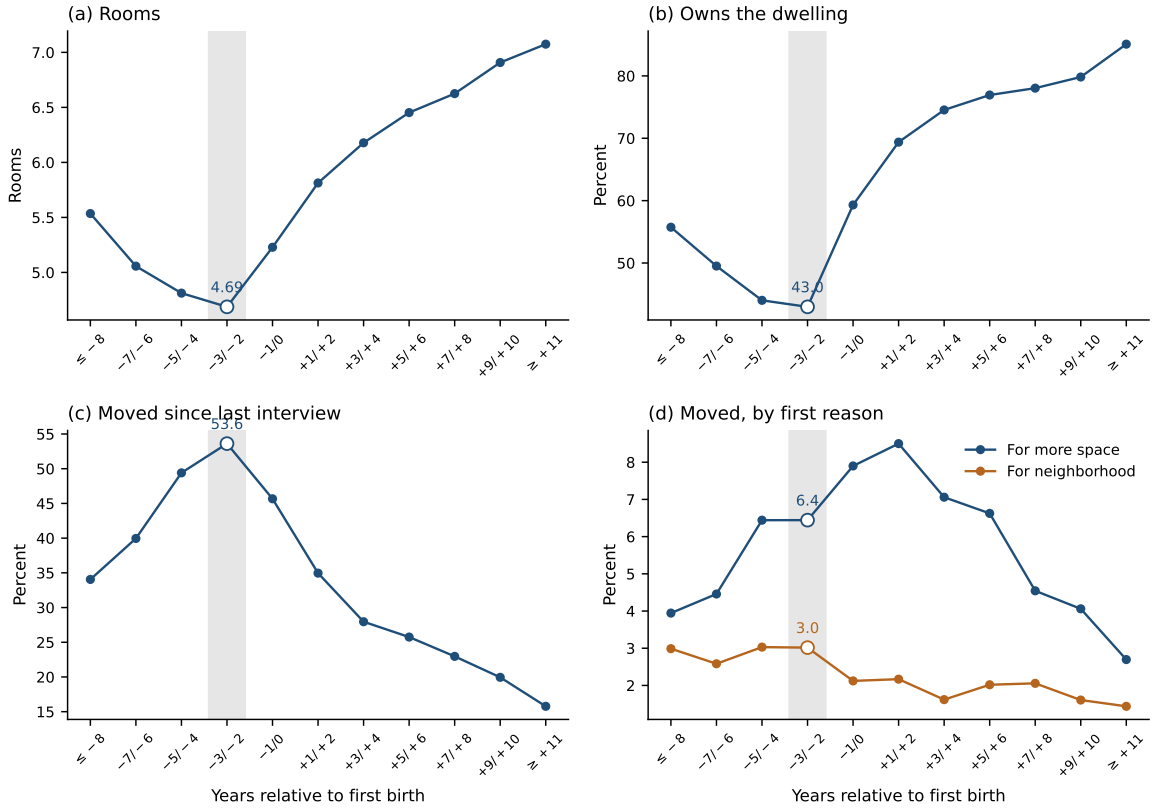
Note: This figure shows the change in rooms around the first birth for two samples: women selected as reference person or spouse in each interview, and treated adults selected as reference person or spouse before birth, with eligible childless controls. Both paths use the full sample for their definition and omit years −3 and −2. The baseline-head path precedes the common-support restriction used for the calibration target.

Figure 24: Ownership and Moving Around the First Birth, Two Samples (PSID)



Note: This figure shows the change in ownership and moving for the two sample definitions of Figure 23, using available observations for each outcome. Reasons for moving in these panels are measured among all observed households, rather than only among movers.

Figure 25: Housing Outcomes by Years Relative to the First Birth (PSID)



Note: This figure shows the levels of the housing outcomes among first-time parents by two-year window of years relative to the first birth, weighted by the PSID longitudinal weight. The sample is the one of panel B of Table 8: adults, women and men, who were reference person or spouse two to three years before the birth. Each outcome uses its available observations. The hollow marker and the shaded band mark the reference window, years −3 and −2, and the label gives its level. Levels are raw means, without fixed effects or controls; before the reference window, these adults need not be reference person or spouse.

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